

Density of Forces Producing Navier–Stokes Blowup

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Abstract

We study the density of smooth external forces for which the three-dimensional Navier–Stokes equations lose classical regularity by a prescribed time $T > 0$. Starting from the compact forced blowup solution of OpenAI, we construct a singular solution near every regular reference solution while preserving its initial velocity. A smooth cutoff of a local vector potential makes the reference velocity vanish near the support of a rescaled blowup solution. The two velocities then have no nonlinear interaction, and the resulting force remains smooth through the blowup time. For fixed viscosity and zero initial velocity, such forces are dense in the relative $L_t^1 H_x^s$ topology on both $\mathbb{T}^3$ and $\mathbb{R}^3$ exactly when $s < 1/2$. On $\mathbb{R}^3$, the corresponding $L_t^2 H_x^s$ threshold is $s < -1/2$. Density below these thresholds holds for every fixed admissible smooth initial velocity. Separate critical estimates give the regular neighborhoods needed for non-density. We also obtain strong approximation of regular reference velocities in energy and dissipation. On $\mathbb{R}^3$, smooth compact forces producing breakdown by T are dense in the completed force spaces $L_t^q H_x^s$ for $q \in \{1, 2\}$ and $s < 2/q - 3/2$, and in $L_t^2 \dot{H}_x^{-1}$, for every fixed admissible smooth initial velocity. Around any whole-space reference regular through T , velocity and force cell averages can be preserved for every $0 \leq t < T$ on any prescribed finite family of Cartesian grids.

Keywords: forced Navier–Stokes equations; finite-time blowup; force-space density; Sobolev spaces; compact support.

1 Introduction

The three-dimensional incompressible Navier–Stokes equations with viscosity $\nu > 0$ and external force f are

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = a. \quad (1)$$

Here u is the velocity, p is the pressure, and a is the prescribed initial velocity. We work on $D = \mathbb{R}^3$ or on the unit torus $D = \mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$, with volume one and periodic velocity and pressure in the latter case. Our starting point is the following theorem, stated with the notation of OpenAI [8, Theorem 1.1].

Theorem 1.1 (OpenAI). *For every $\nu > 0$ there exist a force $f \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3)$, a compact set $K \subset \mathbb{R}^3$, and smooth velocity and pressure fields u, p on $\mathbb{R}^3 \times [0, 1)$ satisfying*

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = 0, \quad (2)$$

such that $\text{supp } u(\cdot, t) \cup \text{supp } p(\cdot, t) \subset K$ for every $0 \leq t < 1$,

$$\sup_{0 \leq t < 1} \|u(t)\|_{L^2(\mathbb{R}^3)} < \infty, \quad \limsup_{t \uparrow 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty. \quad (3)$$

Consequently, there is no smooth solution (u, P) on $\mathbb{R}^3 \times [0, \infty)$ with the same force and initial datum whose kinetic energy is uniformly bounded $\sup_{t \geq 0} \frac{1}{2} \int_{\mathbb{R}^3} |u(x, t)|^2 dx < \infty$.

Fix a viscosity $\nu > 0$, an initial velocity a , and a time $T > 0$. Our aim is to determine in which norms smooth forces producing breakdown of the classical Navier–Stokes solution by time T are dense among smooth forces. The force is allowed to vary, while the initial velocity is kept fixed. This formulation asks how closely a regular flow can be approximated by a flow that becomes singular, when closeness is measured in a prescribed force norm.

Our construction follows a local insertion strategy closely related to the work of Enciso, Peñafiel-Tomás and Peralta-Salas [2] on the incompressible Euler equations. Their Theorem 1.5 constructs weak solutions that agree with a prescribed smooth flow outside a chosen region and have singular behavior inside it. Lemma 2.11 and Section 11 combine local modification of the background, preservation of incompressibility, and insertion of localized singular solutions. In particular, Section 11.3 first makes the background spatially constant in a small region before inserting the singular construction. Our argument uses the same general strategy, with the compact classical forced Navier–Stokes solution of Theorem 1.1 as its input. Here the smooth gluing defect is absorbed into a modified external force; in [2], convex integration removes the Reynolds stress to produce unforced weak Euler solutions. We study the resulting quantitative force-density consequences, preserving the initial velocity and earlier history and identifying the Sobolev thresholds stated below.

In Theorem 1.1, C_c^∞ denotes smooth functions with compact support in the indicated open set, so the force is smooth through the blowup time and vanishes near $t = 0$. For the construction below, fix one solution from Theorem 1.1 and denote its velocity, pressure, and force by (U, P, F) . This distinguishes it from the reference solution to be modified. The corresponding compact solution and its support and zero-data properties are also recorded in the source repository [9].

We now define the spaces in which the force approximation is measured. For background on Fourier transforms, distributions, and Sobolev spaces, see Taylor [13, Chapter 3, Sections 3–4, and Chapter 4, Sections 1 and 3]; the definitions and normalizations used here are specified explicitly below.¹ For $s \in \mathbb{R}$ and $D \in \{\mathbb{R}^3, \mathbb{T}^3\}$, $H^s(D)$ denotes the inhomogeneous L^2 -based Sobolev space of order s . Its norm is

$$\begin{aligned} \|z\|_{H^s(\mathbb{T}^3)}^2 &= \sum_{k \in \mathbb{Z}^3} (1 + 4\pi^2|k|^2)^s |\widehat{z}(k)|^2, \\ \|z\|_{H^s(\mathbb{R}^3)}^2 &= \int_{\mathbb{R}^3} (1 + |\xi|^2)^s |\widehat{z}(\xi)|^2 d\xi, \end{aligned}$$

where $\widehat{z}(k) = \int_{\mathbb{T}^3} z(x) e^{-2\pi i k \cdot x} dx$ on the torus and $\widehat{z}(\xi) = (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-i x \cdot \xi} z(x) dx$ on the whole space. More precisely,

$$\begin{aligned} H^s(\mathbb{R}^3) &= \{z \in \mathcal{S}'(\mathbb{R}^3) : (1 + |\xi|^2)^{s/2} \widehat{z} \in L^2(\mathbb{R}^3)\}, \\ H^s(\mathbb{T}^3) &= \{z \in \mathcal{D}'(\mathbb{T}^3) : \|z\|_{H^s(\mathbb{T}^3)} < \infty\}. \end{aligned}$$

¹On $\mathbb{R}^3$, our Fourier normalization and inhomogeneous Sobolev norm agree with Taylor's. His Bessel multiplier Λ^s corresponds to our $J^s = (I - \Delta)^{s/2}$, whereas we use $\Lambda^s = (-\Delta)^{s/2}$. His torus formulas use 2π -periodic angular coordinates; we use the unit torus. The weights $(1 + |k|^2)^s$ and $(1 + 4\pi^2|k|^2)^s$ give equivalent, not identical, periodic norms after the coordinate change.

Here $\mathcal{S}'(\mathbb{R}^3)$ is the space of tempered distributions and $\mathcal{D}'(\mathbb{T}^3)$ the space of periodic distributions; Fourier transforms and coefficients are understood distributionally. Thus $H^0(D) = L^2(D)$, and negative orders allow distributions that need not be L^2 functions. For vectors and tensors we sum the squared component norms. The notation $\|z\|_p$ means the spatial $L^p(D)$ norm. The homogeneous norm $\dot{H}^s$ replaces the weights above by $(2\pi|k|)^{2s}$ or $|\xi|^{2s}$, respectively. On the torus we omit $k = 0$ and require zero mean; on $\mathbb{R}^3$ the homogeneous norm is used when finite, with the relevant distributional realizations specified in Section 2 and Appendix B. For background on homogeneous Sobolev norms, see Tao [10, Appendix A].²

For a time interval I and a spatial normed space X , $L^q(I; X)$ is the space of strongly measurable maps with finite norm; for the Bochner-space framework, see Hunter [7, Appendix 6.A, Definitions 6.14 and 6.27].³

$$\|z\|_{L^q(I; X)} = \left(\int_I \|z(t)\|_X^q dt \right)^{1/q}, \quad \|z\|_{L^\infty(I; X)} = \operatorname{ess\,sup}_{t \in I} \|z(t)\|_X, \quad (4)$$

where the first formula applies for $1 \leq q < \infty$. In particular,

$$L_t^q H_x^s = L^q(0, \infty; H^s(D)), \quad L_t^q \dot{H}_x^s = L^q(0, \infty; \dot{H}^s(D)), \quad L_t^q L_x^p = L^q(0, \infty; L^p(D))$$

for forces: the time exponent controls integrability, and the Sobolev order measures only spatial regularity. We give each smooth force class the relative topology induced by the stated norm. This differs from the test-function topology, which controls all derivatives on a common compact support. Unless an interval is displayed, force norms use $(0, \infty)$ and velocity norms before blowup use $(0, T)$. For a fixed finite time $T > 0$, the velocity approximation is measured by

$$\|z\|_{E_T} = \|z\|_{L^\infty(0, T; L^2(D))} + \|\nabla z\|_{L^2(0, T; L^2(D))}. \quad (5)$$

Since $\|z\|_{L_t^2 L_x^2} \leq T^{1/2} \|z\|_{L_t^\infty L_x^2}$, this is equivalent to the sum norm on $L_t^\infty L_x^2 \cap L_t^2 H_x^1$ on a fixed finite interval; it imposes no endpoint value at T .

The construction combines compact support, parabolic scaling, and smooth cutoffs. Scaling places the singular solution in an arbitrarily small spatial ball and a short interval ending at T . To insert it into a regular reference solution v , one must also remove v near that ball: direct addition would create nonlinear interaction terms involving the singular velocity. The cutoff construction is the smooth form of the Urysohn lemma: a function is chosen to equal one near a compact set and zero outside a slightly larger open set. Incompressibility requires a further step. Locally write $v = \nabla \times A$ and set

$$w_\varepsilon = -\nabla \times (\eta_\varepsilon \theta_\varepsilon A), \quad u_\varepsilon = v + w_\varepsilon + U_\varepsilon,$$

where U_ε is the rescaled singular solution. The cutoffs are chosen so that $v + w_\varepsilon$ vanishes on a neighborhood of its active support. Both nonlinear interaction terms then vanish identically. The new force is the sum of the original force, the rescaled smooth force, and a smooth correction determined entirely by v and the cutoffs.

This construction explains the positive density results. A force rescales with amplitude ε^{-3} , spatial volume ε^3 , and time scale ε^2 . Its homogeneous $L_t^q \dot{H}_x^s$ scaling factor, whenever that norm is finite, is

$$\varepsilon^{2/q - 3/2 - s}.$$

²Tao uses the unnormalized Fourier transform and includes a compensating factor $(2\pi)^{-3/2}$ in the Fourier formula for the Sobolev norm on $\mathbb{R}^3$; the resulting norm agrees with ours wherever finite. This reference does not replace our explicit conventions for low frequencies and homogeneous-space realizations.

³Hunter states these definitions for Banach-valued functions on a finite interval $(0, T)$. Here the same measurability condition and norm formulas are used on the indicated interval I , including $(0, \infty)$; functions equal almost everywhere are identified.

The background correction has one additional power of ε . On the whole space, negative orders require a separate estimate near frequency zero. On the torus, a localization estimate compares a compactly supported function with its periodization. These arguments give the following thresholds for breakdown from rest:

Domain	Force topology	Density holds exactly when
$\mathbb{T}^3$	$L^1(0, \infty; H^s)$	$s < 1/2$
$\mathbb{R}^3$	$L^1(0, \infty; H^s)$	$s < 1/2$
$\mathbb{R}^3$	$L^2(0, \infty; H^s)$	$s < -1/2$

Theorems 3.1 and 4.1 give the precise statements. Below each threshold, density holds for every fixed admissible smooth initial velocity. The converse statements are proved for initial velocity zero, by constructing a nonempty open set of regular forces. In particular, they require estimates beyond the scaling calculation.

The approximation is also strong at the level of the velocity. Its $L_t^\infty L_x^2$ difference and the $L_{t,x}^2$ norm of its gradient both tend to zero. The solution agrees with the reference until shortly before T , and its speed becomes unbounded exactly at T whenever the reference is regular through that time. For density around a general force, the conclusion is breakdown *by* T : a force whose solution already breaks down earlier is left unchanged.

A related force-density result was proved by Hofmanová, Zhu and Zhu [6, Theorem 1.4, Corollary 4.6, and Theorem 5.1]. In the deterministic setting, forces yielding nonunique Leray–Hopf solutions from rest are dense in $L^1(0, 1; L^2(\mathbb{R}^3))$, and each divergence-free L^2 initial velocity admits a force yielding nonuniqueness. Here the property under study is classical breakdown with a globally smooth force. The compact blowup theorem supplies that property, while the local construction above places it near an arbitrary regular reference. Fujita and Kato [4] and Tao [12] provide background for the critical velocity framework. We use the classical forced local theory of Tao [11, Theorems 5.1 and 5.4] and the classical Sobolev embeddings cited in the appendices. The precise continuation criterion, normalization reductions, and critical force estimates needed here are justified below.

Section 2 specifies the smooth data classes, pressure conventions, and local theory, and derives an energy estimate for the compact singular solution. Section 3 proves the periodic result and its further consequences. Section 4 treats the whole space, including rapidly decaying forces, completed force spaces, and cell averages. Section 5 discusses the scope of the conclusions. Appendices A and B state the classical local theory and Sobolev embeddings, with the continuation argument and the normalization and homogeneous-space details needed on both domains.

2 Definitions and preliminary results

2.1 Smooth data and solution classes

We use the equation, domains, and norms introduced in Section 1. All fields are real, and forces need not be divergence free. We write $L_\sigma^2(D)$ for the divergence-free subspace of $L^2(D; \mathbb{R}^3)$. The allowed initial velocities and forces are

$$\mathcal{X} = \mathcal{X}_{\mathbb{T}^3} = C_{\text{div}}^\infty(\mathbb{T}^3; \mathbb{R}^3), \quad \mathcal{F} = \mathcal{F}_{\mathbb{T}^3} = C_c^\infty(\mathbb{T}^3 \times (0, \infty); \mathbb{R}^3), \quad (6)$$

$$\mathcal{X}_{\mathbb{R}} = H^\infty(\mathbb{R}^3; \mathbb{R}^3) \cap L_\sigma^2(\mathbb{R}^3), \quad (7)$$

where $H^\infty = \bigcap_{m=0}^\infty H^m$ has its usual Fréchet topology. On the whole space set

$$\mathcal{F}_{\mathbb{R}} = \left\{ f \in C^\infty([0, \infty); H^\infty) : \|f\|_{L_t^1 H_x^m} + \|f\|_{L_t^2 H_x^m} < \infty \text{ for every integer } m \geq 0 \right\}. \quad (8)$$

Smoothness into H^∞ means smoothness into each H^m , with one-sided time derivatives at zero. Thus periodic forces vanish near zero and outside a compact time interval; whole-space forces may be nonzero at zero and need not have compact support. Both classes consist of forces defined through any possible singular time of the velocity.

On the torus, velocity and pressure are periodic and $\int_{\mathbb{T}^3} p = 0$. On $\mathbb{R}^3$, a classical velocity belongs to $C([0, S]; H^m)$ for every integer $m \geq 0$ on each compact interval of its lifespan. Its pressure gradient is specified below, and the scalar pressure is determined up to a function of time. In either case local uniqueness defines a maximal classical lifespan, denoted by $T_{\max}^\nu(a, f)$ on the torus and $T_{\max, \mathbb{R}}^\nu(a, f)$ on the whole space. A reference solution is *regular through T* if it extends smoothly to $[0, T + \delta]$ for some $\delta > 0$ in the relevant class.

For fixed $\nu, T > 0$, define

$$\begin{aligned} \mathcal{B}_{\nu, a, T} &= \{f \in \mathcal{F} : T_{\max}^\nu(a, f) \leq T\}, & \mathcal{B}_{\nu, T}^0 &= \mathcal{B}_{\nu, 0, T}, \\ \mathcal{B}_{\nu, a, T}^{\mathbb{R}} &= \{f \in \mathcal{F}_{\mathbb{R}} : T_{\max, \mathbb{R}}^\nu(a, f) \leq T\}. \end{aligned} \quad (9)$$

We call these the forces producing classical breakdown by T . The inserted solutions satisfy the more specific condition of unbounded speed at their terminal time.

2.2 Homogeneous spaces and Fourier multipliers

Write $\Lambda = (-\Delta)^{1/2}$ and $J = (I - \Delta)^{1/2}$, with the Fourier conventions of Section 1. On mean-zero periodic fields, the homogeneous and inhomogeneous Sobolev norms are equivalent at each fixed order. Appendix B specifies the whole-space homogeneous realization for the positive orders used in the embeddings. For the completed energy-force space we fix

$$\dot{H}^{-1}(\mathbb{R}^3) = \{h \in \mathcal{S}' : \widehat{h} = F \text{ is a measurable function and } |\xi|^{-1}F \in L^2\}. \quad (11)$$

Functions representing distributions are identified almost everywhere. The map $h \mapsto |\xi|^{-1}\widehat{h}$ is an isometric bijection onto L^2 : its inverse sends G to the inverse transform of $|\xi|G$, which is tempered because

$$\left| \int |\xi| G(\xi) \phi(\xi) \, d\xi \right| \leq \|G\|_2 \| |\xi| \phi \|_2 \quad (\phi \in \mathcal{S}).$$

Thus this is a separable Hilbert space with no polynomial ambiguity. The same statement for H^s follows from the isometry $h \mapsto \langle \xi \rangle^s \widehat{h}$. Real vector fields correspond to the closed subspace with $F(-\xi) = \overline{F(\xi)}$.

2.3 Projection, pressure, and local existence

The Leray projection $\mathbb{P}$ has Fourier symbol $I - \xi \otimes \xi / |\xi|^2$ on $\mathbb{R}^3 \setminus \{0\}$ and the same symbol at $k \neq 0$ on the torus. At the periodic zero mode it is the identity. It is bounded on every H^s and commutes with derivatives and the heat semigroup. The projected equation is

$$\partial_t u - \nu \Delta u = -\mathbb{P} \nabla \cdot (u \otimes u) + \mathbb{P} f. \quad (12)$$

On the torus, pressure is recovered from

$$\Delta p = \nabla \cdot f - \sum_{i,j} \partial_i \partial_j (u_i u_j), \quad \int_{\mathbb{T}^3} p = 0.$$

The right side has zero mean. On $\mathbb{R}^3$ we require

$$\nabla p = (I - \mathbb{P})(f - \nabla \cdot (u \otimes u)) =: G. \quad (13)$$

For smooth H^∞ data, G is smooth and its Fourier transform is parallel to ξ . Hence $\partial_j G_k = \partial_k G_j$, and a potential is

$$p(x, t) = \int_0^1 G(rx, t) \cdot x \, dr.$$

Differentiating and using this symmetry gives $\partial_j p = \int_0^1 \partial_r [r G_j(rx, t)] \, dr = G_j(x, t)$. There is no requirement that $p \in L^2(\mathbb{R}^3)$. Its gradient belongs to L^2 , so a nonzero constant pressure gradient is excluded. On the torus, the requirement of periodic pressure also excludes an affine pressure.

Proposition 2.1 (Local existence, uniqueness, and continuation). *For each initial velocity in the stated class and each force smooth into every H^m on compact time intervals, equation (1) has a unique maximal smooth velocity, with pressure determined as above. If a solution is defined on $[0, S)$, $S < \infty$, and*

$$\int_0^S \|u(t)\|_{H^2(D)}^2 \, dt < \infty, \quad (14)$$

then it extends smoothly beyond S .

Appendix A derives this statement from the classical forced local theory, with one common existence interval for all Sobolev orders, and proves the displayed continuation criterion. It explains the viscosity and pressure conventions and the periodic mean reduction; no compact support of the whole-space velocity is required.

2.4 Energy and initial vanishing of the compact solution

We use the fixed solution (U, P, F) chosen after Theorem 1.1. Its energy estimate supplies the time-integrated dissipation bound and the initial vanishing needed for rescaling.

Lemma 2.2 (Energy, dissipation and initial vanishing). *The localized solution satisfies*

$$M := \sup_{0 \leq t < 1} \|U(t)\|_2 < \infty, \quad D := \|\nabla U\|_{L^2((0,1) \times \mathbb{R}^3)} < \infty.$$

Both U and P vanish on an initial time interval and therefore extend smoothly by zero to negative times.

Proof. On each closed interval $[0, b]$ with $b < 1$, compact spatial support justifies the energy identity

$$\frac{1}{2}(\|U(t)\|_2^2)' + \nu \|\nabla U(t)\|_2^2 = \langle F(t), U(t) \rangle.$$

Set $N(t) = \int_0^t \|F(s)\|_2 \, ds$. Dropping dissipation gives $\frac{1}{2}(\|U\|_2^2)' \leq \|F\|_2 \|U\|_2$. For $\zeta > 0$, the derivative of $(\|U\|_2^2 + \zeta^2)^{1/2}$ is at most $\|F\|_2$. Integrating from zero and letting $\zeta \downarrow 0$ proves $\|U(t)\|_2 \leq N(t)$.

Substituting this bound into the integrated energy identity gives

$$\|U(t)\|_2^2 + 2\nu \int_0^t \|\nabla U(s)\|_2^2 \, ds \leq 2 \int_0^t \|F(s)\|_2 N(s) \, ds = N(t)^2. \quad (15)$$

Since F is smooth and time compact, the right side is bounded independently of $t < 1$. Monotone convergence of the dissipation integral proves $D < \infty$.

The temporal support of F is a compact subset of $(0, \infty)$, so $F = 0$ on $[0, \tau]$ for some $\tau > 0$. Estimate (15) gives $U = 0$ there. Equation (2) then implies $\nabla P = 0$, and compact spatial support fixes this spatial constant to zero. Vanishing on an interval proves smoothness of both zero extensions. $\square$

3 The torus

All spatial norms in this section are on $\mathbb{T}^3$, unless another domain is shown. The spaces $\mathcal{X}$, $\mathcal{F}$, and $\mathcal{B}_{\nu,a,T}$ are those defined in Section 2.

3.1 Statement of the density theorem

Theorem 3.1 (Sobolev density threshold). *Equip $\mathcal{F}$ with its relative $L^1(0, \infty; H^s(\mathbb{T}^3))$ norm topology.*

- (i) *For every fixed $a \in \mathcal{X}$, the set $\mathcal{B}_{\nu,a,T}$ is dense if $s < 1/2$.*
- (ii) *For zero initial velocity,*

$$\mathcal{B}_{\nu,T}^0 \text{ is dense in } \mathcal{F} \iff s < \frac{1}{2}.$$

We first construct a singular solution close to a regular reference. The force estimates yield density below the stated threshold. A separate small-force estimate then proves non-density at and above it.

3.2 Localization and scaling

Lemma 3.2 (Uniform localization of fractional norms). *Let B be a fixed coordinate ball whose closure lies inside a torus fundamental cube. If z is smooth and supported in B , let $z_{\mathbb{R}}$ be its zero extension in that cube to $\mathbb{R}^3$, and let $z_{\mathbb{T}^3}$ be its periodization. For $0 < s < 1$,*

$$\|z_{\mathbb{T}^3}\|_{H^s(\mathbb{T}^3)} \leq C_{s,B} (\|z_{\mathbb{R}}\|_2 + \|z_{\mathbb{R}}\|_{\dot{H}^s(\mathbb{R}^3)}). \quad (16)$$

The constant is uniform as the support shrinks inside B . At $s = 0$ the L^2 norms are equal; at $s = 1$ the corresponding equality holds for the L^2 gradient norms.

Proof. We first identify the difference expressions with the Fourier norms used in this article. Define

$$c_s = \int_{\mathbb{R}^3} \frac{|e^{ih_1} - 1|^2}{|h|^{3+2s}} dh.$$

It is positive and finite: near zero the numerator is at most $|h|^2$, giving a radial integral $\int_0^1 r^{1-2s} dr$; away from zero it is at most four, giving $\int_1^\infty r^{-1-2s} dr$. For a compact smooth Z on $\mathbb{R}^3$, change variables $y = x + h$, then use Plancherel and Tonelli on the nonnegative integrands to obtain

$$\begin{aligned} I_{\mathbb{R}}(Z) &:= \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} \frac{|Z(x+h) - Z(x)|^2}{|h|^{3+2s}} dx dh \\ &= \int_{\mathbb{R}^3} |\widehat{Z}(\xi)|^2 \left(\int_{\mathbb{R}^3} \frac{|e^{i\xi \cdot h} - 1|^2}{|h|^{3+2s}} dh \right) d\xi = c_s \|Z\|_{\dot{H}^s(\mathbb{R}^3)}^2. \end{aligned}$$

For $\xi \neq 0$, rotation and dilation identify the inner integral with $c_s |\xi|^{2s}$; it is zero when $\xi = 0$.

Let Q be the chosen unit fundamental cube and define the nonnegative periodic kernel, off its lattice singularities, by

$$K_s(h) = \sum_{n \in \mathbb{Z}^3} |h + n|^{-3-2s}.$$

For a smooth periodic z , partitioning $\mathbb{R}^3$ into translates of Q and using periodicity gives

$$\begin{aligned} I_{\mathbb{T}^3}(z) &:= \int_Q \int_Q |z(x) - z(y)|^2 K_s(x - y) \, dx \, dy \\ &= \int_{\mathbb{R}^3} \frac{\|z(\cdot + h) - z\|_{L^2(Q)}^2}{|h|^{3+2s}} \, dh \\ &= \sum_{\ell \in \mathbb{Z}^3} |\widehat{z}(\ell)|^2 \int_{\mathbb{R}^3} \frac{|e^{2\pi i \ell \cdot h} - 1|^2}{|h|^{3+2s}} \, dh = c_s \|z - \widehat{z}(0)\|_{\dot{H}^s(\mathbb{T}^3)}^2. \end{aligned}$$

The second line uses ordinary periodic L^2 norms. Parseval proves the third line and Tonelli justifies interchanging the nonnegative sum and integral. Thus both identifications have the same explicitly specified constant c_s , with the frequency factor 2π retained on the torus. Also $(1 + r^2)^s \leq 1 + r^{2s}$ for $0 < s < 1$, so

$$\|z\|_{H^s(\mathbb{T}^3)}^2 \leq \|z\|_2^2 + c_s^{-1} I_{\mathbb{T}^3}(z).$$

It remains to compare the two difference integrals uniformly. Set $d = \text{dist}(\overline{B}, \partial Q) > 0$. For $x \in B$, $y \in Q$ and $n \neq 0$ we have $|x - y + n| \geq d$: the point $y - n$ lies outside Q up to its boundary. For $|n| > 2\sqrt{3}$ the same distance is at least $|n|/2$. Consequently

$$\sup_{x \in B, y \in Q, n \neq 0} |x - y + n|^{-3-2s} \leq C_{s,d} < \infty,$$

and the identical estimate holds with x, y reversed. The $n = 0$ term of $I_{\mathbb{T}^3}(z_{\mathbb{T}^3})$ is at most $I_{\mathbb{R}}(z_{\mathbb{R}})$. For all the other terms use $|z(x) - z(y)|^2 \leq 2|z(x)|^2 + 2|z(y)|^2$; each nonzero term on the right has its corresponding point in B . As $|Q| = 1$, their total is at most $4C_{s,d} \|z_{\mathbb{R}}\|_2^2$. This proves (16). Both d and all constants refer to B, Q, s , independently of any smaller support. At orders zero and one, integration over the single copy of the support proves the stated L^2 and gradient identities. $\square$

Choose a compact set K_* containing K and the spatial projection of $\text{supp } F$. Fix $x_0 \in B$, where B is the ball in Lemma 3.2. For sufficiently small $\varepsilon > 0$, require

$$2\varepsilon^2 < T, \quad x_0 + \varepsilon K_* \subset B, \quad t_\varepsilon = T - \varepsilon^2.$$

We extend F by zero to nonpositive source times; this extension is smooth because its temporal support is compact in $(0, \infty)$. Using this extension and the negative-time zero extensions of U, P from Lemma 2.2, define

$$\begin{aligned} U_\varepsilon(x, t) &= \varepsilon^{-1} U\left(\frac{x - x_0}{\varepsilon}, \frac{t - t_\varepsilon}{\varepsilon^2}\right), \\ P_\varepsilon(x, t) &= \varepsilon^{-2} P\left(\frac{x - x_0}{\varepsilon}, \frac{t - t_\varepsilon}{\varepsilon^2}\right), \\ F_\varepsilon(x, t) &= \varepsilon^{-3} F\left(\frac{x - x_0}{\varepsilon}, \frac{t - t_\varepsilon}{\varepsilon^2}\right). \end{aligned} \tag{17}$$

The first two fields are used for $t < T$; the force is defined for all positive times. Place the rescaled whole-space solution in B and then periodize it. The torus contains a single copy of its support.

Proposition 3.3 (Scaling at fixed viscosity). *The fields in (17) solve the periodic momentum equation at viscosity ν , start from zero, and have unbounded speed at T . A spatially constant pressure adjustment enforces the mean-zero normalization. Their norms satisfy*

$$\|U_\varepsilon\|_{L^\infty(0,T;L^2)} = \varepsilon^{1/2}M, \quad \|\nabla U_\varepsilon\|_{L^2(0,T;L^2)} = \varepsilon^{1/2}D, \quad (18)$$

$$\|F_\varepsilon\|_{L^q(0,\infty;L^p)} = \varepsilon^{\alpha(p,q)} \|F\|_{L^q(0,\infty;L^p)}, \quad \alpha(p,q) = -3 + \frac{3}{p} + \frac{2}{q}, \quad (19)$$

for $1 \leq p, q \leq \infty$. For $0 \leq s \leq 1$,

$$\|F_\varepsilon\|_{L^1(0,\infty;H^s(\mathbb{T}^3))} \leq C_s(\varepsilon^{1/2} + \varepsilon^{1/2-s}). \quad (20)$$

In particular the force tends to zero in $L_t^1 H_x^s$ for every $s < 1/2$, including negative s .

Proof. Writing $y = (x - x_0)/\varepsilon$ and $\sigma = (t - t_\varepsilon)/\varepsilon^2$, every term of the momentum equation gains the common factor ε^{-3} . Incompressibility is preserved and no viscosity rescaling occurs. The temporal extension is smooth by initial vanishing. The speed identity $\|U_\varepsilon(t)\|_\infty = \varepsilon^{-1} \|U(\sigma)\|_\infty$ proves unboundedness as $t \uparrow T$.

At a fixed time the velocity has amplitude factor ε^{-1} and spatial volume factor ε^3 , giving the first identity in (18). Its gradient has amplitude factor ε^{-2} . Squaring and integrating in both space and time gives $\varepsilon^{-4}\varepsilon^3\varepsilon^2 = \varepsilon$ times the original dissipation integral, proving the second identity.

For finite p, q , the force's spatial norm gains $\varepsilon^{-3+3/p}$ and its time norm gains $\varepsilon^{2/q}$. The positive-time domain includes the whole transformed support because $t_\varepsilon > 0$ and the original force vanishes at negative times. This proves (19); the essential-supremum cases follow by the same change of variables with zero reciprocal exponents.

Euclidean homogeneous Sobolev scaling gives, for $0 < s < 1$,

$$\left\| \varepsilon^{-3} F((\cdot - x_0)/\varepsilon, \sigma) \right\|_{\dot{H}^s(\mathbb{R}^3)} = \varepsilon^{-3/2-s} \|F(\cdot, \sigma)\|_{\dot{H}^s(\mathbb{R}^3)}.$$

One may verify the exponent either by Fourier transform or by the double integral in Lemma 3.2: its squared scaling factor is $\varepsilon^{-6}\varepsilon^6\varepsilon^{-3-2s} = \varepsilon^{-3-2s}$. Time integration contributes ε^2 . The inhomogeneous L^2 term has order $\varepsilon^{1/2}$. Lemma 3.2 therefore proves (20). At $s = 0$ use the mixed-norm identity, and at $s = 1$ use the gradient scaling. For $s < 0$, the Fourier definition gives $\|z\|_{H^s} \leq \|z\|_2$, so the $s = 0$ estimate proves convergence without any homogeneous negative-order claim. $\square$

The force amplitude is $\varepsilon^{-3} \|F\|_\infty$, even though its integrated norm tends to zero below the critical order. The homogeneous $L_t^1 \dot{H}_x^{1/2}$ norm is scale invariant. Section 3.5 supplies the regularity estimate needed at that order.

3.3 Insertion into a regular solution

Fix a reference solution (v, π, g) smooth on $[0, T + \delta]$, where $\delta > 0$, $g \in \mathcal{F}$, and $v(0) = a \in \mathcal{X}$. We first make the reference velocity vanish near the support of U_ε . This prevents nonlinear interaction terms from introducing a singularity into the new force.

The smooth cutoff functions used below can be constructed directly. If a compact set K_* lies in an open set O , choose $d > 0$ so that its closed $3d$ -neighborhood lies in O . Convolve the indicator of its $2d$ -neighborhood with a nonnegative smooth mollifier supported in the ball of radius d and of integral one. The resulting function takes values in $[0, 1]$, equals one near K_* , and has compact support in O . This is the smooth cutoff form of the Urysohn lemma. The same construction in one dimension provides the time cutoff.

Lemma 3.4 (A local divergence-free cutoff). *Let v be smooth and divergence free in a spatial ball centered at x_0 . In that ball, with $y = x - x_0$, define*

$$A(x, t) = \int_0^1 r v(x_0 + ry, t) \times y \, dr. \quad (21)$$

Then $\nabla \times A = v$. Choose $\theta \in C_c^\infty(\mathbb{R}^3)$ equal to one on a neighborhood of K_ , and choose $\eta \in C_c^\infty((-2, 2))$ equal to one on $[-1, 1]$. Put*

$$\theta_\varepsilon(x) = \theta((x - x_0)/\varepsilon), \quad \eta_\varepsilon(t) = \eta((t - T)/\varepsilon^2), \quad w_\varepsilon = -\nabla \times (\eta_\varepsilon \theta_\varepsilon A). \quad (22)$$

For sufficiently small ε , w_ε is a smooth, divergence-free field supported inside the coordinate ball and in $(T - 2\varepsilon^2, T + 2\varepsilon^2)$. During the active interval before blowup of U_ε ,

$$v + w_\varepsilon = 0 \quad \text{on an open neighborhood of } \text{supp } U_\varepsilon(t). \quad (23)$$

Proof. Use the vector identity

$$\nabla \times (V \times y) = V \nabla \cdot y - y \nabla \cdot V + (y \cdot \nabla)V - (V \cdot \nabla)y.$$

For $V(y) = v(x_0 + ry, t)$, incompressibility gives $\nabla_y \cdot V = 0$, $\nabla_y \cdot y = 3$, and $(V \cdot \nabla)y = V$. Consequently,

$$\begin{aligned} \nabla_y \times (r v(x_0 + ry, t) \times y) &= 2r v(x_0 + ry, t) + r^2 (y \cdot \nabla_x) v(x_0 + ry, t) \\ &= \frac{d}{dr} (r^2 v(x_0 + ry, t)). \end{aligned}$$

Integrating from $r = 0$ to $r = 1$ proves $\nabla \times A = v$.

Take ε small enough that the scaled support of θ is strictly inside the coordinate ball and $2\varepsilon^2 < \min(T, \delta)$. Since the spatial cutoff is zero near the boundary of the coordinate ball, extending (22) by zero gives a globally smooth periodic field. Its divergence vanishes because it is a curl. The temporal support follows directly from η_ε .

The rescaled velocity U_ε is zero before $t_\varepsilon = T - \varepsilon^2$. For $t_\varepsilon \leq t < T$, the variable $(t - T)/\varepsilon^2$ belongs to $[-1, 0)$, on which $\eta = 1$. On a neighborhood of the localized solution support, $\theta_\varepsilon = 1$ and its derivatives vanish. There $w_\varepsilon = -\nabla \times A = -v$, proving (23). $\square$

Lemma 3.5 (Bounds for the background correction). *For w_ε in Lemma 3.4, define*

$$H_\varepsilon = \partial_t w_\varepsilon - \nu \Delta w_\varepsilon + (v \cdot \nabla) w_\varepsilon + (w_\varepsilon \cdot \nabla) v + (w_\varepsilon \cdot \nabla) w_\varepsilon. \quad (24)$$

The force H_ε is smooth across T and extends by zero to all times outside its cutoff support. Its spatial support has volume $O(\varepsilon^3)$, and its temporal support has length $O(\varepsilon^2)$. For every spatial multi-index β and integer $j \geq 0$,

$$|\partial_t^j \partial_x^\beta w_\varepsilon| \leq C_{\beta,j} \varepsilon^{-2j-|\beta|}, \quad |\partial_x^\beta H_\varepsilon| \leq C_\beta \varepsilon^{-2-|\beta|}. \quad (25)$$

In particular,

$$\|w_\varepsilon\|_{E_T} \leq C \varepsilon^{3/2}, \quad (26)$$

$$\|H_\varepsilon\|_{L^q(0, \infty; L^p)} \leq C_{p,q} \varepsilon^{-2+3/p+2/q} = C_{p,q} \varepsilon^{\alpha(p,q)+1}, \quad (27)$$

$$\|H_\varepsilon\|_{L^1(0, \infty; H^s)} \leq C_s (\varepsilon^{3/2} + \varepsilon^{3/2-s}) \quad (0 \leq s \leq 1). \quad (28)$$

All constants are independent of sufficiently small ε .

Proof. Introduce $z = (x - x_0)/\varepsilon$ and $\sigma = (t - T)/\varepsilon^2$. Then

$$A(x_0 + \varepsilon z, T + \varepsilon^2 \sigma) = \varepsilon \mathcal{A}_\varepsilon(z, \sigma),$$

where

$$\mathcal{A}_\varepsilon(z, \sigma) = \int_0^1 r v(x_0 + \varepsilon r z, T + \varepsilon^2 \sigma) \times z \, dr.$$

On the fixed support of $\theta(z)\eta(\sigma)$, every fixed-order derivative of $\mathcal{A}_\varepsilon$ is bounded uniformly in ε . Differentiation of v in z or σ introduces nonnegative powers of ε , and the reference is smooth on a fixed neighborhood of the relevant spacetime set. Thus

$$w_\varepsilon(x_0 + \varepsilon z, T + \varepsilon^2 \sigma) = W_\varepsilon(z, \sigma) := -\nabla_z \times (\eta(\sigma)\theta(z)\mathcal{A}_\varepsilon(z, \sigma))$$

is a uniformly smooth family supported in a fixed cylinder. This proves the first bound in (25).

To check the force bound, set $V_\varepsilon(z, \sigma) = v(x_0 + \varepsilon z, T + \varepsilon^2 \sigma)$. Formula (24) becomes

$$\begin{aligned} H_\varepsilon(x_0 + \varepsilon z, T + \varepsilon^2 \sigma) &= \varepsilon^{-2} [\partial_\sigma W_\varepsilon - \nu \Delta_z W_\varepsilon + \varepsilon (V_\varepsilon \cdot \nabla_z) W_\varepsilon \\ &\quad + \varepsilon^2 (W_\varepsilon \cdot \nabla_x) v(x_0 + \varepsilon z, T + \varepsilon^2 \sigma) + \varepsilon (W_\varepsilon \cdot \nabla_z) W_\varepsilon]. \end{aligned}$$

The bracketed expression and each of its spatial derivatives are uniformly bounded on the fixed cylinder, proving the second bound in (25). In particular, H_ε depends only on the smooth reference and the cutoffs.

From the support volume and duration,

$$\|w_\varepsilon\|_{L_t^\infty L_x^2} \leq C\varepsilon^{3/2}, \quad \|\nabla w_\varepsilon\|_{L_t^2 L_x^2} \leq C\varepsilon^{-1}\varepsilon^{3/2}\varepsilon = C\varepsilon^{3/2}.$$

This proves (26). The amplitude bound $C\varepsilon^{-2}$ gives (27) by the same support calculation, including the essential-supremum endpoints.

For (28), the rescaled force profiles have uniformly bounded spatial H^s norms, supported on a fixed spatial set during a fixed interval of σ . An amplitude ε^{-2} gives the spatial homogeneous factor $\varepsilon^{-1/2-s}$; time integration adds ε^2 . Thus the homogeneous contribution is $O(\varepsilon^{3/2-s})$, while the L^2 contribution is $O(\varepsilon^{3/2})$. Lemma 3.2 transfers these estimates to the torus. Smooth zero extension in time is valid because η , with all its derivatives, vanishes near the fixed interval endpoints $\sigma = -2$ and $\sigma = 2$, where the reference remains smooth. $\square$

Theorem 3.6 (Exact local insertion). *Let (v, π, g) be a periodic solution smooth on $[0, T + \delta]$, with $\delta > 0$, $g \in \mathcal{F}$ and $v(0) = a \in \mathcal{X}$. Fix any nonempty coordinate ball. For all sufficiently small $\varepsilon > 0$, there are $g_\varepsilon \in \mathcal{F}$ and a solution u_ε with*

- (i) $T_{\max}^\nu(a, g_\varepsilon) = T$ and $\limsup_{t \uparrow T} \|u_\varepsilon(t)\|_\infty = \infty$;
- (ii) $u_\varepsilon = v$ for $0 \leq t \leq T - 2\varepsilon^2$;
- (iii) $u_\varepsilon - v$ divergence free and supported, for every $t < T$, in a ball of diameter $O(\varepsilon)$ inside the chosen ball;
- (iv) the simultaneous bounds

$$\|u_\varepsilon - v\|_{E_T} \leq (M + D)\varepsilon^{1/2} + C\varepsilon^{3/2}, \tag{29}$$

$$\|g_\varepsilon - g\|_{L_t^q L_x^p} \leq C_{p,q}(\varepsilon^{\alpha(p,q)} + \varepsilon^{\alpha(p,q)+1}), \tag{30}$$

$$\|g_\varepsilon - g\|_{L_t^1 H_x^s} \leq C_s(\varepsilon^{1/2-s} + \varepsilon^{3/2-s}) \quad (0 \leq s < 1/2). \tag{31}$$

Here force norms are taken over $(0, \infty)$, and $\alpha(p, q) = -3 + 3/p + 2/q$ for $1 \leq p, q \leq \infty$. For $s < 0$, the force difference also tends to zero in $L_t^1 H_x^s$.

Proof. Choose the rescaled singular solution and the correction from Lemmas 3.4–3.5, with all supports inside the prescribed ball, and set

$$u_\varepsilon = v + w_\varepsilon + U_\varepsilon, \quad p_\varepsilon = \pi + P_\varepsilon, \quad g_\varepsilon = g + H_\varepsilon + F_\varepsilon. \quad (32)$$

Initially use the compact representative P_ε ; subtract its spatial mean to normalize p_ε .

Let $b_\varepsilon = v + w_\varepsilon$. Expanding the reference equation and (24) gives

$$\partial_t b_\varepsilon + (b_\varepsilon \cdot \nabla) b_\varepsilon - \nu \Delta b_\varepsilon + \nabla \pi = g + H_\varepsilon.$$

Adding the localized solution equation produces the equation for u_ε except for

$$(b_\varepsilon \cdot \nabla) U_\varepsilon + (U_\varepsilon \cdot \nabla) b_\varepsilon.$$

Both terms are identically zero. On a neighborhood of $\text{supp } U_\varepsilon$ this follows from $b_\varepsilon = 0$, including its derivatives, by (23). Outside that support U_ε and its derivatives vanish, including at the support boundary by smoothness. Before its starting time it vanishes everywhere. Thus (1) holds exactly for (32) on $[0, T)$.

Each summand of the velocity is divergence free. The correction is zero for $t \leq T - 2\varepsilon^2$, and the localized solution is zero before $T - \varepsilon^2$, so the initial datum and the stated earlier history are unchanged. On the active localized solution support, $u_\varepsilon = U_\varepsilon$. Hence

$$\|u_\varepsilon(t)\|_\infty \geq \|U_\varepsilon(t)\|_\infty$$

and the maximum norm is unbounded as $t \uparrow T$.

The velocity is smooth on every closed interval before blowup. The force F_ε is globally smooth by its original definition, and H_ε is globally smooth by Lemma 3.5. Their time supports remain compact subsets of $(0, \infty)$ when ε is sufficiently small. Therefore $g_\varepsilon \in \mathcal{F}$, independently of the absence of a classical extension of u_ε at T . Uniqueness in Proposition 2.1 identifies the constructed velocity with the maximal solution for (a, g_ε) up to every $T' < T$. Its lifespan is at least T , and the unbounded maximum norm excludes extension beyond T . Thus the lifespan is exactly T .

Finally, the triangle inequality, Proposition 3.3 and Lemma 3.5 give (29)–(31). In (31), the lower-order inhomogeneous terms are absorbed because $0 < \varepsilon \leq 1$ and $s \geq 0$. For negative s , use $\|z\|_{H^s} \leq \|z\|_2$ and the $s = 0$ estimate. $\square$

3.4 Density below the critical order

Proposition 3.7 (Density for each fixed initial velocity). *For $a \in \mathcal{X}$, $\nu > 0$, $T > 0$ and $s < 1/2$,*

$$\forall g \in \mathcal{F} \ \forall \rho > 0 \ \exists f \in \mathcal{F} : \quad \|f - g\|_{L_t^1 H_x^s} < \rho, \quad T_{\max}^\nu(a, f) \leq T.$$

Proof. Fix g and ρ . There are two exhaustive cases.

If $T_{\max}^\nu(a, g) \leq T$, choose $f = g$. It already belongs to $\mathcal{B}_{\nu, a, T}$ and the norm difference is zero.

If $T_{\max}^\nu(a, g) > T$, choose $\delta > 0$ so that $T + \delta < T_{\max}^\nu(a, g)$, and use its smooth solution as the reference in Theorem 3.6. For every sufficiently small ε , that theorem supplies $g_\varepsilon \in \mathcal{B}_{\nu, a, T}$ with lifespan exactly T . For $0 \leq s < 1/2$, (31) tends to zero. For $s < 0$, its $s = 0$ case and the embedding $L^2 \hookrightarrow H^s$ do so. Choose ε so that the force difference is less than ρ and set $f = g_\varepsilon$.

Both choices preserve a , ν , and T , and give the required approximation. $\square$

3.5 Critical regularity and completion of the proof

To prove non-density at $s = 1/2$, we exhibit an open ball of forces whose solutions from rest are globally regular. The following proposition is a forced adaptation of the classical critical small-data theory of Fujita and Kato [4]; see also Tao [12, Theorem 45] for the periodic small-data global existence theorem without external forcing. It expresses persistence of global regularity near the zero solution under a small force in $L_t^1 H_x^{1/2}$. We include a proof for this precise force class, allowing a nonzero spatial mean: the argument removes the mean, controls the critical velocity norm, and applies the continuation criterion. This estimate is independent of the compact blowup theorem used in the density construction.

Proposition 3.8 (Global regularity for small critical force). *There is $c > 0$, depending only on the unit torus and the stated norm conventions, such that*

$$g \in \mathcal{F}, \quad \rho := \|g\|_{L^1(0,\infty;H^{1/2})} < c\nu \quad (33)$$

imply $T_{\max}^\nu(0, g) = \infty$.

Proof. Let u be the maximal classical solution from Proposition 2.1. All estimates are first made on a compact subinterval of its lifespan.

Removal of the spatial mean. Define

$$\bar{g}(t) = \int_{\mathbb{T}^3} g(x, t) \, dx, \quad m(t) = \int_{\mathbb{T}^3} u(x, t) \, dx, \quad v = u - m, \quad h = g - \bar{g}.$$

Integrating the periodic equation eliminates the convective, Laplacian and pressure terms, giving $m' = \bar{g}$ and $m(0) = 0$. Thus

$$|m(t)| \leq \int_0^t |\bar{g}(s)| \, ds \leq \rho. \quad (34)$$

The mean-zero field v solves

$$\partial_t v + (v \cdot \nabla)v + (m \cdot \nabla)v - \nu \Delta v + \nabla p = h. \quad (35)$$

For any real Sobolev order, the spatially constant transport $m(t) \cdot \nabla$ commutes with the corresponding Fourier multiplier and is skew-adjoint in L^2 . It therefore contributes zero to both energy identities used below. This step is necessary because the force is not assumed to have zero spatial mean.

Critical energy estimate. Let $\Lambda = (-\Delta)^{1/2}$ on mean-zero fields, and set

$$y(t) = \|\Lambda^{1/2} v(t)\|_2, \quad z(t) = \|\Lambda^{3/2} v(t)\|_2, \quad b(t) = \|h(t)\|_{\dot{H}^{1/2}}.$$

Testing (35) against Λv gives

$$\frac{1}{2}(y^2)' + \nu z^2 = -\langle (v \cdot \nabla)v, \Lambda v \rangle + \langle h, \Lambda v \rangle.$$

The pressure term is zero because $\nabla \cdot \Lambda v = 0$. By Lemma A.1 and Hölder's inequality,

$$|\langle (v \cdot \nabla)v, \Lambda v \rangle| \leq \|v\|_3 \|\nabla v\|_3 \|\Lambda v\|_3 \leq C_0 y z^2.$$

The force term is bounded by

$$|\langle h, \Lambda v \rangle| = |\langle \Lambda^{1/2} h, \Lambda^{1/2} v \rangle| \leq b(t) y(t).$$

Consequently,

$$\frac{1}{2}(y^2)' + (\nu - C_0 y) z^2 \leq b y. \quad (36)$$

Removing the zero Fourier mode cannot increase the relevant norm, so

$$\int_0^\infty b(t) dt \leq \rho. \quad (37)$$

A continuity argument. On any interval where $y \leq \nu/(2C_0)$, (36) implies $\frac{1}{2}(y^2)' \leq by$. For $\zeta > 0$,

$$\frac{d}{dt}(y^2 + \zeta^2)^{1/2} \leq b \frac{y}{(y^2 + \zeta^2)^{1/2}} \leq b.$$

Since $y(0) = 0$, integration and $\zeta \downarrow 0$ yield

$$y(t) \leq \int_0^t b(s) ds \leq \rho \quad (38)$$

throughout such an interval. Choose $c < 1/(4C_0)$. If a first time with $y = \nu/(2C_0)$ existed, continuity would permit the preceding calculation up to that time, where it would give the strictly smaller bound $y \leq \rho < \nu/(4C_0)$. Thus the bound holds throughout the classical lifespan.

In particular the critical dissipation also obeys

$$\frac{1}{2}y(t)^2 + \frac{\nu}{2} \int_0^t z(s)^2 ds \leq \int_0^t b(s)y(s) ds \leq \rho^2.$$

We next obtain a stronger integrability bound that directly implies classical continuation.

The H^1 estimate. Take the inner product of (35) with $-\Delta v$. The pressure and mean-transport terms vanish. The nonlinear term satisfies

$$\begin{aligned} |\langle (v \cdot \nabla)v, \Delta v \rangle| &\leq \|v\|_3 \|\nabla v\|_6 \|\Delta v\|_2 \\ &\leq C_1 y \|\Delta v\|_2^2. \end{aligned}$$

Here $\|\nabla v\|_6 \leq C \|\Delta v\|_2$ follows from Lemma B.1, applied to each mean-zero derivative: $\|\nabla(\partial_j v)\|_2 \leq \|\Delta v\|_2$ by its Fourier series. Choose also $c < 1/(4C_1)$. By (38), the convection term is at most $(\nu/4) \|\Delta v\|_2^2$. Young's inequality bounds the force term by $(\nu/4) \|\Delta v\|_2^2 + C\nu^{-1} \|h\|_2^2$. After multiplying the resulting inequality by two and weakening its positive coefficient if necessary,

$$(\|\nabla v\|_2^2)' + \nu \|\Delta v\|_2^2 \leq C\nu^{-1} \|h\|_2^2. \quad (39)$$

Since $v(0) = 0$, integration gives a uniform bound on $\|\nabla v(t)\|_2^2$ and on $\int_0^t \|\Delta v(s)\|_2^2 ds$. The integral of $\|h\|_2^2$ over all positive times is finite because each fixed force is smooth and time compact. It need not be small.

Continuation. Suppose the maximal lifespan were a finite number S . For mean-zero v , Fourier series give $\|v\|_{H^2} \leq C \|\Delta v\|_2$. Together with (39) and (34), the orthogonality of the constant and mean-zero Fourier modes gives

$$\begin{aligned} \int_0^S \|u(t)\|_{H^2}^2 dt &= \int_0^S (|m(t)|^2 + \|v(t)\|_{H^2}^2) dt \\ &\leq S\rho^2 + C\nu^{-2} \int_0^\infty \|h(t)\|_2^2 dt < \infty. \end{aligned}$$

Proposition 2.1 then extends u beyond S , a contradiction. Thus the maximal lifespan is infinite. The choice of c uses only the constants in the periodic Sobolev estimates, not the amplitude, derivatives or support size of a particular force. $\square$

Corollary 3.9 (Non-density at and above the critical exponent). *For every $s \geq 1/2$ and $T > 0$, $\mathcal{B}_{\nu,T}^0$ is not dense in $\mathcal{F}$ with the relative $L_t^1 H_x^s$ topology.*

Proof. At $s = 1/2$, the set

$$\{g \in \mathcal{F} : \|g\|_{L_t^1 H_x^{1/2}} < c\nu\}$$

is a nonempty relative open ball, containing zero, and is disjoint from $\mathcal{B}_{\nu,T}^0$ by Proposition 3.8. For $s \geq 1/2$, the Fourier weights imply $\|g(t)\|_{H^{1/2}} \leq \|g(t)\|_{H^s}$. The ball $\|g\|_{L_t^1 H_x^s} < c\nu$ is therefore also disjoint from the singular-force set. A dense subset cannot miss a nonempty open set. $\square$

Proof of Theorem 3.1. Proposition 3.7 gives density for every fixed smooth initial velocity when $s < 1/2$. Corollary 3.9 gives non-density for zero initial velocity when $s \geq 1/2$. These are the two assertions of the theorem. $\square$

The converse has been proved for zero initial velocity. The behavior for general nonzero initial velocities at or above the critical order remains outside this classification.

3.6 Other force norms, trajectory approximation, and data pairs

Corollary 3.10 (A sufficient mixed-norm region). *For every fixed $a \in \mathcal{X}$, the set $\mathcal{B}_{\nu,a,T}$ is dense in $\mathcal{F}$ with its relative $L^q(0, \infty; L^p)$ topology whenever*

$$1 \leq p, q \leq \infty, \quad \frac{3}{p} + \frac{2}{q} > 3. \quad (40)$$

Proof. Use the same two cases as in Proposition 3.7. In the regular-reference case, (30) tends to zero because $\alpha(p, q) > 0$ and hence $\alpha(p, q) + 1 > 0$. $\square$

The sufficient region includes $L_t^1 L_x^2$ and $L_t^2 L_x^{4/3}$. The other mixed Lebesgue topologies require further analysis.

Corollary 3.11 (Strong closure in energy and dissipation). *Fix $a \in \mathcal{X}$. Let $\mathcal{R}_{a,T}$ be the trajectories with initial velocity a and forces in $\mathcal{F}$ that are smooth through T . Let $\mathcal{S}_{a,T}$ be the trajectories with the same initial velocity and forces in $\mathcal{F}$ that are smooth on $[0, T)$, have finite E_T norm, and have unbounded maximum velocity at T . Then*

$$\mathcal{R}_{a,T} \subseteq \overline{\mathcal{S}_{a,T}}^{E_T}. \quad (41)$$

For each reference pair (v, g) and every fixed $s < 1/2$, the approximating pairs can be chosen with

$$(u_\varepsilon, g_\varepsilon) \longrightarrow (v, g) \quad \text{in } E_T \times L^1(0, \infty; H^s).$$

Proof. A reference smooth through T is smooth through $T + \delta$ for some $\delta > 0$, by local continuation; equivalently one may take this as the meaning of smoothness through the endpoint. Apply Theorem 3.6. Its energy estimate gives convergence in E_T , and its force estimates give simultaneous convergence. The reference has finite E_T norm and the localized solution and correction do also, so u_ε belongs to the specified ambient energy space.

To verify the claimed interpretation of the norm, note that

$$\|z\|_{L^2(0,T;L^2)} \leq T^{1/2} \|z\|_{L^\infty(0,T;L^2)}.$$

Thus E_T controls both terms of $L_t^2 H_x^1$, while the usual sum norm controls E_T directly. This proves the equivalence used in (41). No endpoint value or classical continuation of u_ε is implied by convergence in this norm. $\square$

Proposition 3.12 (Projection of extended singular data). *Define*

$$\mathfrak{B}_{\nu,T} = \{(a, f) \in \mathcal{X} \times \mathcal{F} : T_{\max}^\nu(a, f) \leq T\}.$$

If $s < 1/2$, this set is dense in the product of any topology on $\mathcal{X}$ and the relative $L_t^1 H_x^s$ topology on $\mathcal{F}$. Its projection onto $\mathcal{X}$ is all of $\mathcal{X}$. The family obtained while requiring $a = 0$ projects instead to the singleton $\{0\}$.

Proof. Take a point (a, g) and a basic product neighborhood $U \times V$ containing it. The initial datum a already belongs to U . Proposition 3.7 supplies a force $f \in V$ with $T_{\max}^\nu(a, f) \leq T$. Hence $(a, f) \in \mathfrak{B}_{\nu,T} \cap (U \times V)$, proving density. Applying the same proposition with any fixed g and any positive radius gives at least one singular force over each a , which proves the projection statement. The final assertion follows directly from fixing the initial velocity to zero. $\square$

The projection statement has the quantifier order $\forall a \exists f$, not $\exists f \forall a$. For a fixed prescribed force f_* , the set

$$\{a \in \mathcal{X} : T_{\max}^\nu(a, f_*) \leq T\}$$

is a different object and is not classified by these force-changing constructions. In a nontrivial normed initial-velocity space, $\{0\}$ is not dense: if $a \neq 0$, the ball of radius $\|a\|/2$ centered at a excludes zero.

There is also a distinction between the two terminal-time assertions. Theorem 3.6 gives singularity exactly at T around every reference regular through T . Proposition 3.7 uses breakdown by T , because a general reference may already break down earlier and is then left unchanged. It does not prove density of exactly-time- T singular forces around every earlier-breaking reference.

Remark 3.13 (Concentration and force amplitudes). The force convergence in Theorem 3.6 is compatible with diverging pointwise amplitudes. Since the original force is nonzero,

$$\|g_\varepsilon - g\|_{L_{t,x}^\infty} \geq \varepsilon^{-3} \|F\|_\infty - C\varepsilon^{-2} \longrightarrow \infty.$$

Here the first term is the amplitude of F_ε and the second bounds H_ε . The force F is nonzero by (15), since U blows up. Hence the approximating smooth forces have unbounded amplitudes and cannot obey uniform bounds on all derivatives.

3.7 Insertion in a bounded domain

For the following bounded-domain statement, $H^s(\Omega)$ is the space of restrictions of $H^s(\mathbb{R}^3)$ distributions, with quotient norm

$$\|z\|_{H^s(\Omega)} = \inf_{Z|_\Omega = z} \|Z\|_{H^s(\mathbb{R}^3)}. \quad (42)$$

This convention applies also to negative orders. At order zero it is the usual $L^2(\Omega)$ norm. For every fixed compact $K \Subset \Omega$ and $s \in \mathbb{R}$, smooth fields supported in K obey

$$\|z\|_{H^s(\Omega)} \leq \|E_0 z\|_{H^s(\mathbb{R}^3)} \leq C_{s,K,\Omega} \|z\|_{H^s(\Omega)}, \quad \text{supp } z \subset K, \quad (43)$$

where $E_0 z$ is extension by zero. To prove the second inequality, fix $\chi \in C_c^\infty(\Omega)$ equal to one near K . For every extension Z of z , $\chi Z = E_0 z$. Multiplication by this fixed χ is bounded on $H^s(\mathbb{R}^3)$ for every real s : the Fourier convolution formula and

$$\frac{(1 + |\xi|^2)^{s/2}}{(1 + |\eta|^2)^{s/2}} \leq C_s (1 + |\xi - \eta|^2)^{|s|/2}$$

reduce the estimate to Young's $L^1 * L^2 \rightarrow L^2$ inequality, since $\hat{\chi}$ decays faster than every power. Taking the infimum over Z proves the bound; the first inequality follows from the definition. The constant depends on the fixed cutoff and s , not on a smaller support or a shrinking scale ε . Applying the pointwise bounds in time gives the same comparison for all the mixed norms in (4), provided K is fixed for all times. No bounded zero-extension operator on arbitrary $H^s(\Omega)$ is asserted. All vector and tensor norms in these definitions are the square root of the sum of the squared component norms.

Corollary 3.14 (Interior no-slip insertion). *Let $\Omega \subset \mathbb{R}^3$ be a bounded box or a bounded smooth domain. Suppose that, for some $\delta > 0$, (v, π, g) is a smooth no-slip solution on $[0, T + \delta]$, and that g is smooth on $\bar{\Omega} \times [0, \infty)$ with temporal support compact in $(0, \infty)$. Here smoothness on a closed spacetime slab means restriction of a C^∞ field from an open neighborhood of that slab; this convention also specifies smoothness at the edges and corners of a box. We assume v, π have this regularity on $\bar{\Omega} \times [0, T + \delta]$, and g on each finite closed slab. The equation and incompressibility hold in Ω , and $v|_{\partial\Omega} = 0$. Pressure may be normalized by zero spatial mean. Existence of this compatible reference is an assumption of the corollary. The construction in Theorem 3.6 applies inside any prescribed interior ball, preserving the initial velocity and no-slip boundary values. The energy estimates use $L^2(\Omega)$ and the force estimates use $L^1(0, \infty; H^s(\Omega))$, with the restriction norm (42). For every real s , the domain and zero-extended force-difference norms are comparable by (43), with a constant independent of ε ; the convergence assertion remains $s < 1/2$.*

Proof. The vector potential and all cutoff estimates are local. Choose the correction and localized solution supports in a smaller closed ball strictly inside Ω . The computation of the momentum equation in Theorem 3.6 is unchanged. The new velocity equals the reference in a fixed boundary collar for all times before blowup, so the no-slip boundary values are preserved. The force correction is supported in the same interior region and has the same globally smooth time extension.

For $0 \leq s < 1/2$, the Euclidean scaling proof directly bounds the zero extensions of the force differences in $L^1(0, \infty; H^s(\mathbb{R}^3))$. For $s < 0$, use $\|E_0(g_\varepsilon - g)\|_{H^s(\mathbb{R}^3)} \leq \|E_0(g_\varepsilon - g)\|_{L^2(\mathbb{R}^3)}$ at each time. All these force-difference supports, including those after T , lie in one fixed compact interior ball K for every sufficiently small ε . Consequently (43) gives the asserted domain estimates and the scale-independent comparison after time integration. The energy estimates follow by integrating over that ball. Finally, classical uniqueness on a no-slip domain follows from the same difference-energy calculation as in Proposition 2.1; its boundary terms vanish. Thus the constructed solution is singular exactly at T . $\square$

3.8 Further constructions

Proposition 3.15 (Infinite-dimensional variations away from the endpoints). *Fix the localized solution (U, P, F) of Theorem 1.1, and fix a spacetime cylinder $Q = B_0 \times (\tau_0, \tau_1)$ with $0 < \tau_0 < \tau_1 < 1$ and $B_0 \Subset \mathbb{R}^3$ an open ball. For any $b \in C_c^\infty(Q; \mathbb{R}^3)$ with $\nabla \cdot b = 0$, define*

$$\begin{aligned}\tilde{U} &= U + b, & \tilde{P} &= P, \\ \tilde{F} &= F + \partial_t b - \nu \Delta b + (U \cdot \nabla)b + (b \cdot \nabla)U + (b \cdot \nabla)b.\end{aligned}\tag{44}$$

These fields give an infinite-dimensional affine family of distinct velocities with zero initial data, finite energy and dissipation, and the same late singular behavior as U . For each fixed nonzero b , the family obtained by replacing b with λb is non-isolated as $\lambda \rightarrow 0$ in every fixed smooth seminorm of the velocity and force differences on their compact support.

Proof. Expanding the momentum equation for $U + b$ gives exactly the force in (44); its divergence remains zero. Every correction term is supported in $\text{supp } b$, a compact subset of Q . The original

singular solution is smooth with bounded derivatives of every fixed order on a neighborhood of that set. Thus the force correction extends smoothly by zero to the whole positive time axis, including time one, despite the singularity of U there.

Since $b = 0$ near time zero and near time one, $\tilde{U}(0) = 0$ and $\tilde{U} = U$ on a fixed late interval. The same unbounded-speed limit holds. The energy and dissipation norms of b are finite by compact smooth support, so the triangle inequality gives the corresponding bounds for $\tilde{U}$.

To verify infinite dimensionality explicitly, choose countably many disjoint small balls inside B_0 . In each ball choose a smooth compactly supported vector potential with nonzero curl, and multiply that curl by a fixed nonzero time bump in (τ_0, τ_1) . These divergence-free fields are linearly independent because their spatial supports are disjoint. Their finite linear combinations belong to the allowed class. The map $b \mapsto U + b$ is injective and affine, so its image is infinite dimensional.

For the non-isolation statement, the velocity difference is λb , while the force difference has the form

$$\lambda L_U b + \lambda^2 (b \cdot \nabla) b, \quad L_U b = \partial_t b - \nu \Delta b + (U \cdot \nabla) b + (b \cdot \nabla) U.$$

On the fixed compact support, all coefficients and their derivatives are bounded. Hence for each integer $m \geq 0$ the C^m norm of the force difference is at most $C_m |\lambda| + C'_m \lambda^2$, and the velocity difference is at most $C''_m |\lambda|$. Both tend to zero. The forces vary in this family; no open singularity basin for one fixed force follows. $\square$

Proposition 3.16 (Finitely many prescribed singular regions). *Fix finitely many disjoint interior balls $B_1, \dots, B_N$ in the torus or in a bounded three-dimensional domain. At fixed $\nu > 0$ and $T > 0$, there is a smooth forced solution starting from rest, smooth on $[0, T)$, with finite energy and dissipation, such that*

$$\limsup_{t \uparrow T} \|u(t)\|_{L^\infty(B_j)} = \infty \quad (1 \leq j \leq N).$$

In a bounded domain its boundary condition is homogeneous no-slip.

Proof. Choose $x_j \in B_j$ and scales $\varepsilon_j > 0$ sufficiently small that the complete scaled compact set $x_j + \varepsilon_j K_*$ lies strictly inside B_j and $\varepsilon_j^2 < T$. Apply (17) to each localized solution with start time $T - \varepsilon_j^2$, and denote the resulting fields by U_j, P_j, F_j . Set

$$u = \sum_{j=1}^N U_j, \quad p = \sum_{j=1}^N P_j, \quad f = \sum_{j=1}^N F_j.$$

For $i \neq j$, the supports of U_i and U_j are separated, so $(U_i \cdot \nabla) U_j = 0$. The sum therefore satisfies the momentum equation exactly. All force supports are compact in positive time and all forces are smooth. Initial vanishing gives $u(0) = 0$.

On B_j the velocity equals U_j , proving the stated separate limsup for each ball. Disjoint supports also give

$$\sup_{t < T} \|u(t)\|_2^2 \leq M^2 \sum_{j=1}^N \varepsilon_j, \quad \int_0^T \|\nabla u(t)\|_2^2 dt = D^2 \sum_{j=1}^N \varepsilon_j.$$

These sums are finite. In a bounded domain, all fields vanish in a boundary collar; hence $u = 0$ on the boundary. On the torus a spatially constant pressure adjustment gives the chosen gauge. $\square$

Only finitely many solutions are superposed, all with terminal time T . The sequences along which their maxima diverge may differ; the conclusion is a separate limsup statement in each ball.

Proposition 3.17 (Conservative forcing from rest). *On the torus, let $f = -\nabla\phi$ for a globally defined periodic scalar ϕ . On a bounded domain, let $f = -\nabla\phi$ and impose homogeneous no-slip velocity. In either case, any smooth solution starting from rest is identically zero on its classical lifespan.*

Proof. Incompressibility and periodicity or the zero normal boundary trace give

$$\int f \cdot u = - \int \nabla\phi \cdot u = 0.$$

The classical energy identity, integrated from zero, becomes

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \, ds = 0.$$

Both terms are nonnegative, so $u(t) = 0$ at every time before blowup. The periodic-potential condition excludes affine potentials, whose gradients need not be removable by the allowed periodic pressure gauge. $\square$

Thus a nontrivial solution starting from rest requires a nonconservative force.

4 The whole space

We now work on $\mathbb{R}^3$, with the data classes $\mathcal{X}_{\mathbb{R}}$ and $\mathcal{F}_{\mathbb{R}}$ from Section 2. The construction is again spatially local. The new issues are the behavior of negative Sobolev norms near frequency zero and the absence of a spectral gap.

4.1 Statement of the density theorem

Theorem 4.1 (Two whole-space Sobolev thresholds). *Fix $\nu, T > 0$, let $q \in \{1, 2\}$, and set $s_q = 2/q - 3/2$. In the relative $L^q(0, \infty; H^s(\mathbb{R}^3))$ topology on $\mathcal{F}_{\mathbb{R}}$:*

- (i) *For every fixed $a \in \mathcal{X}_{\mathbb{R}}$, $\mathcal{B}_{\nu, a, T}^{\mathbb{R}}$ is dense if $s < s_q$.*
- (ii) *For zero initial velocity, $\mathcal{B}_{\nu, 0, T}^{\mathbb{R}}$ is dense if and only if $s < s_q$.*

Thus the thresholds are $1/2$ for $L_t^1 H_x^s$ and $-1/2$ for $L_t^2 H_x^s$. Around every reference regular through T , the approximating solution can have the same initial velocity, the same earlier history, singularity exactly at T , and a velocity difference tending to zero in E_T .

The force classes remain smooth, equipped with the relative norm topology. For $q = 2$, the converse uses a regular neighborhood whose radius depends on T .

4.2 Local construction and negative Sobolev estimates

The compact solution of Theorem 1.1 is already defined on $\mathbb{R}^3$. Use the scaling (17) directly, without periodization. The energy identities (18) and mixed force identity (19) follow from the same changes of variables. The vector potential in Lemma 3.4 is local, so its proof applies in any Euclidean ball. In Lemma 3.5, the support, derivative, energy, and mixed Lebesgue bounds are also local; only the final transfer of fractional norms to the torus is unnecessary. Thus the same w_ε and H_ε are available on $\mathbb{R}^3$, with the same uniform bounds. We record the result and prove the additional negative-order estimates.

Theorem 4.2 (Whole-space local insertion). *Let (v, π, g) be the solution for $a \in \mathcal{X}_{\mathbb{R}}$ and $g \in \mathcal{F}_{\mathbb{R}}$, regular through $T + \delta$ for some $\delta > 0$. Fix any nonempty open ball $B \subset \mathbb{R}^3$. For all sufficiently small $\varepsilon > 0$, there are $g_\varepsilon \in \mathcal{F}_{\mathbb{R}}$ and a solution u_ε such that*

$$T_{\max, \mathbb{R}}^\nu(a, g_\varepsilon) = T, \quad \limsup_{t \uparrow T} \|u_\varepsilon(t)\|_\infty = \infty, \quad u_\varepsilon = v \quad (0 \leq t \leq T - 2\varepsilon^2).$$

The velocity difference is supported inside B at every $t < T$, and $g_\varepsilon - g \in C_c^\infty(B \times (0, \infty))$. Moreover,

$$\|u_\varepsilon - v\|_{E_T} \leq (M + D)\varepsilon^{1/2} + C\varepsilon^{3/2}. \quad (45)$$

For $q \in \{1, 2\}$ the force difference tends to zero in $L_t^q H_x^s$ whenever $s < 2/q - 3/2$. All of these conclusions hold for the same family of inserted solutions.

Proof. Place the fields in (17) inside B , and choose the compact cutoffs of Lemma 3.4. Define $A, w_\varepsilon, H_\varepsilon$ by (21), (22) and (24). These formulas only use v on a fixed compact neighborhood of B and of time T , so the preceding lemmas apply. In particular $v + w_\varepsilon = 0$ on a neighborhood of the active localized solution support. Put

$$u_\varepsilon = v + w_\varepsilon + U_\varepsilon, \quad p_\varepsilon = \pi + P_\varepsilon, \quad g_\varepsilon = g + H_\varepsilon + F_\varepsilon.$$

Writing $b_\varepsilon = v + w_\varepsilon$, expansion of the equation leaves only the cross-advection terms $(b_\varepsilon \cdot \nabla)U_\varepsilon + (U_\varepsilon \cdot \nabla)b_\varepsilon$. They vanish on a neighborhood of the localized solution support because $b_\varepsilon = 0$ there. Outside that support, the localized solution and its derivatives vanish; smoothness gives the same conclusion at the support boundary. When the localized solution is inactive it is identically zero. Thus the two terms vanish pointwise everywhere, and the equation is exact. The pressure difference may be chosen to be the compact scalar P_ε : the equation gives an L^2 pressure gradient, and applying $I - \mathbb{P}$ recovers (13). Both force corrections are globally smooth and spacetime compact, including across T , so their sum preserves membership in $\mathcal{F}_{\mathbb{R}}$.

The initial value and earlier history are unchanged. At each $t < T$ the compact singular solution and cutoff correction lie in H^∞ , and the maximum norm of U_ε is unbounded at T . Proposition 2.1 identifies the solution with the unique maximal solution. An extension through T would be bounded in $C_t H^2$ on a neighborhood of T , hence bounded in L_x^∞ by (54), contradicting the blowup of U_ε . Thus its maximal lifespan is exactly T . Equation (18) and the correction estimate in Lemma 3.5 give (45) by the triangle inequality.

It remains to justify every asserted force topology, including negative orders. Let

$$\beta(q, s) = \frac{2}{q} - \frac{3}{2} - s.$$

For $0 \leq s \leq 1$, Euclidean scaling and $\|z\|_{H^s} \leq C_s(\|z\|_2 + \|z\|_{\dot{H}^s})$ give

$$\begin{aligned} \|F_\varepsilon\|_{L_t^q H_x^s} &\leq C_{q,s}(\varepsilon^{2/q-3/2} + \varepsilon^{\beta(q,s)}), \\ \|H_\varepsilon\|_{L_t^q H_x^s} &\leq C_{q,s}(\varepsilon^{2/q-1/2} + \varepsilon^{\beta(q,s)+1}). \end{aligned} \quad (46)$$

The second line uses the uniformly smooth compact rescaled profile of Lemma 3.5, whose amplitude is ε^{-2} .

For $-3/2 < s < 0$, every smooth compact profile has finite $\dot{H}^s$ norm. Indeed, on $|\xi| < 1$ its Fourier transform is bounded by its L^1 norm and $\int_{|\xi| < 1} |\xi|^{2s} d\xi < \infty$; on $|\xi| \geq 1$ its L^2 norm suffices. These bounds are uniform for the rescaled correction profiles, which have common compact support and uniform derivatives. Since $(1 + |\xi|^2)^s \leq |\xi|^{2s}$, scaling gives

$$\|F_\varepsilon\|_{L_t^q H_x^s} \leq C_{q,s} \varepsilon^{\beta(q,s)}, \quad \|H_\varepsilon\|_{L_t^q H_x^s} \leq C_{q,s} \varepsilon^{\beta(q,s)+1}. \quad (47)$$

The time norms run over the entire positive axis. Each rescaled force has time support of length $O(\varepsilon^2)$, including any part after T .

For $q = 1$, (46) proves convergence for $0 \leq s < 1/2$; all $s < 0$ follow from $\|z\|_{H^s} \leq \|z\|_2$. For $q = 2$, (47) proves convergence for $-3/2 < s < -1/2$. If $s \leq -3/2$, choose $r \in (-3/2, -1/2)$ with $r > s$ and use $\|z\|_{H^s} \leq \|z\|_{H^r}$. This last step avoids making any false homogeneous scaling assertion at indices where a generic compact profile can have an infinite homogeneous norm. $\square$

4.3 The critical estimate for time-integrable forcing

Proposition 4.3 (Global regularity for small critical data and L^1 force). *There is a universal $c > 0$ such that, for $a \in \mathcal{X}_{\mathbb{R}}$ and $f \in \mathcal{F}_{\mathbb{R}}$,*

$$\|a\|_{\dot{H}^{1/2}} + \|f\|_{L_t^1 \dot{H}_x^{1/2}} < c\nu \implies T_{\max, \mathbb{R}}^\nu(a, f) = \infty.$$

In particular, for $a = 0$, the ball $\|f\|_{L_t^1 \dot{H}_x^{1/2}} < c\nu$ consists of globally regular inputs.

Proof. Set $\Lambda = (-\Delta)^{1/2}$, $y = \|\Lambda^{1/2}u\|_2$, $z = \|\Lambda^{3/2}u\|_2$, and $b = \|f\|_{\dot{H}^{1/2}}$. Lemma B.1 in Appendix B gives

$$\|u\|_3 \leq Cy, \quad \|\nabla u\|_3 + \|\Lambda u\|_3 \leq Cz.$$

Testing the projected equation against Λu consequently yields

$$\frac{1}{2}(y^2)' + (\nu - C_0 y)z^2 \leq by. \quad (48)$$

On an interval with $y \leq \nu/(2C_0)$, regularization by $(y^2 + \zeta^2)^{1/2}$ and then $\zeta \downarrow 0$ gives

$$y(t) \leq y(0) + \int_0^t b(s) \, ds.$$

Choosing $c < 1/(4C_0)$ gives the bound throughout the lifespan by continuity, including times at which $y = 0$.

To prove continuation, testing against $-\Delta u$ gives

$$|\langle (u \cdot \nabla)u, \Delta u \rangle| \leq \|u\|_3 \|\nabla u\|_6 \|\Delta u\|_2 \leq C_1 y \|\Delta u\|_2^2.$$

The derivative estimate $\|\nabla u\|_6 \leq C \|\Delta u\|_2$ follows from the $\dot{H}^1 \rightarrow L^6$ case of Lemma B.1, since Plancherel gives $\|D^2 u\|_2 = \|\Delta u\|_2$. Decrease c so that $C_1 y \leq \nu/4$. Young's inequality then gives

$$(\|\nabla u\|_2^2)' + \nu \|\Delta u\|_2^2 \leq C\nu^{-1} \|f\|_2^2. \quad (49)$$

This controls the gradient and the integral of the squared Laplacian. It does not control low frequencies by itself. The separate ordinary energy identity, with regularized norm division, supplies

$$\|u(t)\|_2 \leq \|a\|_2 + \int_0^t \|f(s)\|_2 \, ds =: K(t). \quad (50)$$

For every finite S within or at the maximal lifespan, $K(S) < \infty$, and the Fourier inequality

$$\|u\|_{H^2}^2 \leq C(\|u\|_2^2 + \|\Delta u\|_2^2)$$

therefore gives

$$\int_0^S \|u(t)\|_{H^2}^2 \, dt \leq CSK(S)^2 + C\nu^{-1} \|\nabla a\|_2^2 + C\nu^{-2} \int_0^S \|f(t)\|_2^2 \, dt < \infty.$$

Proposition 2.1 excludes every finite maximal lifespan. The low frequencies have been controlled directly by the ordinary energy estimate. Finally $\|f\|_{\dot{H}^{1/2}} \leq \|f\|_{H^{1/2}}$ proves the stated inhomogeneous consequence. $\square$

4.4 The critical estimate for square-integrable forcing

Proposition 4.4 (Regularity on a prescribed finite interval). *For each $\nu, S > 0$ there is $r_{\nu, S} > 0$ such that*

$$f \in \mathcal{F}_{\mathbb{R}}, \quad \|f\|_{L^2(0, \infty; H^{-1/2})} < r_{\nu, S} \implies T_{\max, \mathbb{R}}^{\nu}(0, f) > S.$$

One can take $r_{\nu, S} = c\nu^{3/2}e^{-C\nu S}$ for suitable universal positive constants c, C .

Proof. Use the inhomogeneous multiplier $J = (I - \Delta)^{1/2}$ and set

$$Y = \|u\|_{H^{1/2}}, \quad Z = \|\nabla u\|_{H^{1/2}}, \quad B = \|f\|_{H^{-1/2}}.$$

Fourier weights give $\|u\|_{H^{3/2}}^2 = Y^2 + Z^2$. Testing the equation against Ju yields an energy identity with dissipation νZ^2 . By Lemma B.1 and Fourier Cauchy–Schwarz,

$$\begin{aligned} |\langle (u \cdot \nabla)u, Ju \rangle| &\leq \|u\|_3 \|\nabla u\|_3 \|Ju\|_3 \leq CYZ(Y^2 + Z^2)^{1/2} \leq C_0 Y(Y^2 + Z^2), \\ |\langle f, Ju \rangle| &\leq B(Y^2 + Z^2)^{1/2} \leq B(Y + Z). \end{aligned}$$

Thus, while $Y \leq \theta\nu$ for a sufficiently small universal $\theta > 0$, Young’s inequality gives constants C_2, C_3 with

$$(Y^2)' + \nu Z^2 \leq C_2 \nu Y^2 + C_3 \nu^{-1} B^2. \quad (51)$$

For instance absorb the $C_0 Y Z^2$ and BZ terms into dissipation, use $C_0 Y^3 \leq C_0 \theta \nu Y^2$, and bound BY by a multiple of $\nu Y^2 + \nu^{-1} B^2$. Since $Y(0) = 0$, Grönwall gives, for $t \leq S$ while $Y \leq \theta\nu$,

$$Y(t)^2 \leq C_3 \nu^{-1} e^{C_2 \nu S} \|f\|_{L_t^2 H_x^{-1/2}}^2.$$

Choose $r_{\nu, S}$ so that the right side is less than $\theta^2 \nu^2 / 4$. The first possible time with $Y = \theta\nu$ would contradict this bound, so the bound holds up to the lesser of S and the maximal lifespan.

Take θ smaller if necessary. Then $\|u\|_3 \leq CY \leq C\theta\nu$ allows the H^1 absorption in (49). Because $f \in \mathcal{F}_{\mathbb{R}}$, its $L_t^1 L_x^2$ and $L_t^2 L_x^2$ norms are finite, even though they are not required to be small. The separate L^2 estimate (50) and the last continuation calculation in Proposition 4.3 exclude a maximal lifespan at or before S .

The radius depends on S . The inhomogeneous norm is essential here because smallness in $H^{-1/2}$ does not control the homogeneous negative norm at low frequencies. $\square$

Proof of Theorem 4.1. Fix a, g and a positive radius in the indicated force norm. There are two cases. If $T_{\max, \mathbb{R}}^{\nu}(a, g) \leq T$, take $f = g$. Otherwise choose $\delta > 0$ so that the reference exists through $T + \delta$, and apply Theorem 4.2. For $s < s_q$, its force difference is smaller than the prescribed radius when ε is sufficiently small, its initial velocity is exactly a , and its lifespan is exactly T . This proves density for each fixed smooth initial velocity.

For zero initial velocity and $q = 1$, Proposition 4.3 provides an open ball about zero disjoint from $\mathcal{B}_{\nu, 0, T}^{\mathbb{R}}$ in $L_t^1 H_x^{1/2}$. The inclusion $H^s \hookrightarrow H^{1/2}$ with norm at most one for $s \geq 1/2$ gives the same obstruction in each stronger metric. For $q = 2$, use Proposition 4.4 with $S = T$ and then $H^s \hookrightarrow H^{-1/2}$ for $s \geq -1/2$. These nonempty relative open balls prove non-density. The energy and earlier-history assertions are part of the insertion theorem. $\square$

4.5 Compact support, rapid decay, and completed force spaces

Define the following two subclasses of $\mathcal{F}_{\mathbb{R}}$:

$$\begin{aligned} \mathcal{F}_c &= C_c^{\infty}(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3), \\ \mathcal{F}_{\text{rd}} &= \left\{ f \in C^{\infty}(\mathbb{R}^3 \times [0, \infty)) : \sup_{x \in \mathbb{R}^3, t \geq 0} (1 + |x| + t)^N |\partial_x^{\alpha} \partial_t^j f(x, t)| < \infty \text{ for all } N, \alpha, j \right\}. \end{aligned}$$

Vector values are understood in the second line. The initial class corresponding to the rapid-decay formulation is $\mathcal{S}_\sigma = \mathcal{S}(\mathbb{R}^3; \mathbb{R}^3) \cap L_\sigma^2$. These are exactly the decay requirements relevant to [3, equations (4)–(5)]; a common numerical bound on their seminorms is not imposed.

Corollary 4.5 (The same thresholds for compact and rapidly decaying forces). *Theorem 4.1 remains valid with $\mathcal{F}_\mathbb{R}$ replaced by $\mathcal{F}_c$ or $\mathcal{F}_{\text{rd}}$. In particular it holds for every fixed $a \in \mathcal{S}_\sigma$ in the rapid-decay class, with the complete if-and-only-if classification at $a = 0$.*

Proof. The initial value is preserved exactly, and the force correction in Theorem 4.2 is spacetime compact. Addition of that correction preserves each stated force class, including all rapid-decay bounds. The two-case density proof therefore stays within the chosen subclass. Each critical regular ball contains zero and remains a nonempty relative open ball in that subclass. The local framework only requires $a \in \mathcal{X}_\mathbb{R}$, so it applies in particular to Schwartz initial data. Nothing here requires the nonzero reference velocity or its pressure to be spatially compact. $\square$

The force spaces in the next proposition are natural for the energy class. With $V = H^1(\mathbb{R}^3; \mathbb{R}^3) \cap L_\sigma^2$, a finite-energy velocity belongs to $L_t^\infty L_\sigma^2 \cap L_t^2 V$. A full distribution in H^{-1} restricts to an element of V' ; the latter records only its action on divergence-free tests. This distinction is discussed in [1, Definition 1 and Remark 4]. Our realization of $\dot{H}^{-1}$ gives the pairing

$$|\langle h, z \rangle| \leq \|h\|_{\dot{H}^{-1}} \|\nabla z\|_2, \quad \left| \int_0^T \langle f, u \rangle \, dt \right| \leq \|f\|_{L_t^2 \dot{H}_x^{-1}} \|\nabla u\|_{L_{t,x}^2}.$$

These inequalities follow by Fourier Cauchy–Schwarz, first for Schwartz functions and then by completion. The inhomogeneous H^{-1} norm instead pairs with H^1 , while $L_t^1 L_x^2$ pairs with $L_t^\infty L_x^2$.

Proposition 4.6 (Density in completed force spaces and strong trajectory closure). *Fix $a \in \mathcal{X}_\mathbb{R}$ and $\nu, T > 0$. Smooth compact forces with $T_{\max, \mathbb{R}}^\nu(a, f) \leq T$ are dense in each full Bochner space $L^q(0, \infty; H^s(\mathbb{R}^3))$ for $q \in \{1, 2\}$ and $s < s_q$. They are also dense in $L^2(0, \infty; \dot{H}^{-1}(\mathbb{R}^3))$.*

For every reference in Theorem 4.2, one may simultaneously arrange

$$\|u_\varepsilon - v\|_{E_T} \longrightarrow 0, \quad \|g_\varepsilon - g\|_{L_t^1 L_x^2} + \|g_\varepsilon - g\|_{L_t^2 H_x^{-1}} + \|g_\varepsilon - g\|_{L_t^2 \dot{H}_x^{-1}} \longrightarrow 0.$$

The homogeneous norm here is a norm of the compact difference; the background itself need not belong to that homogeneous force space.

Proof. We first give the approximation details for the complete spaces just defined. Fix any real s , and let $h \in H^s$. Approximate $G = \langle \xi \rangle^s \widehat{h} \in L^2$ by $G_n \in C_c^\infty$ in L^2 , using truncation and convolution with a smooth approximate identity. Set

$$\widehat{h}_n(\xi) = \langle \xi \rangle^{-s} G_n(\xi).$$

The right side is smooth and compactly supported, so repeated integration by parts in Fourier inversion shows that h_n is Schwartz. By the isometry defining H^s , $h_n \rightarrow h$ in H^s . Now choose $\chi \in C_c^\infty$ equal to one on the unit ball and zero outside the ball of radius two, with $0 \leq \chi \leq 1$, and let $\chi_R(x) = \chi(x/R)$. For fixed n and any nonnegative integer m , Leibniz’ rule shows

$$\|(1 - \chi_R)h_n\|_{H^m} \longrightarrow 0.$$

Indeed, the term with no derivative on the cutoff is a tail of each square-integrable derivative of h_n . Each remaining term has a factor bounded by CR^{-j} , $j \geq 1$, is supported where $|x| \geq R$, and multiplies a Schwartz derivative. Every such L^2 norm tends to zero. Choosing $m \geq \max(s, 0)$ and using $\|z\|_{H^s} \leq \|z\|_{H^m}$ proves spatial compact-smooth density in H^s for every real s .

For $h \in \dot{H}^{-1}$, instead approximate $G = |\xi|^{-1}\widehat{h}$ in L^2 by smooth compactly supported G_n supported away from 0. One first restricts to $1/n < |\xi| < n$, with L^2 error tending to zero, and then smooths each restricted function with a sufficiently small mollification radius and a slightly larger annular cutoff. Set $\widehat{h}_n = |\xi|G_n$. Again h_n is Schwartz and $h_n \rightarrow h$ in $\dot{H}^{-1}$. To make h_n spatially compact, a low/high-frequency split suffices: for every $k \in L^1 \cap L^2$,

$$\begin{aligned} \|k\|_{\dot{H}^{-1}}^2 &= \int_{|\xi|<1} |\xi|^{-2} |\widehat{k}(\xi)|^2 d\xi + \int_{|\xi|\geq 1} |\xi|^{-2} |\widehat{k}(\xi)|^2 d\xi \\ &\leq C \|k\|_1^2 \int_{|\xi|<1} |\xi|^{-2} d\xi + \|k\|_2^2 \leq C' \|k\|_1^2 + \|k\|_2^2. \end{aligned} \quad (52)$$

The integral at the origin is finite in dimension three. Since $(1 - \chi_R)h_n \rightarrow 0$ in both L^1 and L^2 , (52) proves $\chi_R h_n \rightarrow h_n$ in $\dot{H}^{-1}$. A diagonal choice gives compact-smooth density in this realization, without a dual Sobolev-embedding assumption. These constructions also prove the real vector-valued versions: take real parts of each component of the approximants. Complex conjugation is an isometry for every Fourier norm here because its weight is real and even, so taking real parts is a contraction and preserves convergence to real data, smoothness and compact support.

We next make the time and spatial compactness simultaneous. Write X for any of the preceding separable Hilbert spaces and $1 \leq q < \infty$. Given $b \in L^q((0, \infty); X)$, first restrict it to $[1/N, N]$. The omitted L^q norm tends to zero by integrability. On this finite interval, strongly measurable functions can be approximated in L^q by finite simple functions $\sum_{j=1}^J \mathbf{1}_{E_j} b_j$: truncate the norm of b , approximate its values by a countable dense set in X , and retain a finite number of the resulting value sets with arbitrarily small remaining measure. Approximate each b_j by a physical $h_j \in C_c^\infty(\mathbb{R}^3)$ in X . The error is at most

$$\sum_{j=1}^J |E_j|^{1/q} \|b_j - h_j\|_X.$$

Each measurable $E_j \subset [1/N, N]$ can be approximated in measure by a finite union of intervals lying in a fixed compact subset of $(0, \infty)$, by regularity of Lebesgue measure. Smoothing their indicators by convolution gives $\varphi_j \in C_c^\infty((0, \infty))$ with arbitrarily small $\|\mathbf{1}_{E_j} - \varphi_j\|_{L^q}$; the L^q convergence of this smoothing follows first for an interval by estimating its shrinking boundary strips and then for a finite union. The resulting error is at most

$$\sum_{j=1}^J \|\mathbf{1}_{E_j} - \varphi_j\|_{L^q} \|h_j\|_X.$$

Thus the finite sum $\sum_j \varphi_j(t) h_j(x)$ approximates b and is jointly smooth with compact support strictly inside $\mathbb{R}^3 \times (0, \infty)$. This proves the claimed Bochner density with both kinds of compactness, rather than merely compactness of a Sobolev-valued time path.

For the inhomogeneous singular-force density, first choose such a smooth compact force within half the prescribed radius of the target. Corollary 4.5 supplies a singular smooth compact force within the other half. This is density of a smooth subset in the completion, not a definition of classical breakdown for every rough force.

For the homogeneous insertion estimate, apply the Fourier scaling calculation of (47) at $s = -1$. Explicitly, an amplitude ε^{-a} has spatial $\dot{H}^{-1}$ factor $\varepsilon^{5/2-a}$, because $\varepsilon^{-a}\widehat{h}(\cdot/\varepsilon)(\xi) = \varepsilon^{3-a}\widehat{h}(\varepsilon\xi)$. The time L^2 factor is ε . The source force has $a = 3$, while the correction has $a = 2$ and uniformly bounded compact rescaled profiles; their $\dot{H}^{-1}$ norms are uniformly finite by (52). Hence

$$\|F_\varepsilon\|_{L_t^2 \dot{H}_x^{-1}} \leq C\varepsilon^{1/2}, \quad \|H_\varepsilon\|_{L_t^2 \dot{H}_x^{-1}} \leq C\varepsilon^{3/2}.$$

Given a compact smooth reference force, if its lifespan is at most T it already belongs to the required singular set. Otherwise insert the localized solution and use these estimates. Combining this relative approximation with the proved compact-smooth Bochner density establishes density in $L_t^2 \dot{H}_x^{-1}$. Finally these estimates, (45), and the $q = 1, s = 0$ and $q = 2, s = -1$ cases of the insertion estimates prove the simultaneous convergence for the same family. $\square$

As on the torus, every $a \in \mathcal{X}_{\mathbb{R}}$ occurs as the initial velocity of a solution producing breakdown for some smooth force. Thus the projection of the extended singular data onto $\mathcal{X}_{\mathbb{R}}$ is the whole space, with quantifier order $\forall a \exists f$. The preceding approximation concerns finite-energy whole-space solutions and uses one rescaled copy of U in each perturbation.

4.6 Cell averages on prescribed grids

The spaces used for energy estimates should be distinguished from the additional regularity assumed in numerical error estimates. For example, on a bounded no-slip computational domain one uses the solenoidal energy space H with zero normal trace and $V = H_0^1 \cap H$; finite-dimensional velocity and pressure spaces discretize this setting. Guermond, Mineev and Shen [5, Section 2 and Theorem 3.2] state these spaces and the additional smoothness required for their error estimate. The initial pressure is constrained by the equation and boundary conditions. These bounded-domain conventions do not impose boundary conditions at infinity in the whole-space result below.

To relate the continuous norms to a concrete numerical observation, let $\mathcal{T}_h$ be a complete uniform Cartesian grid of $\mathbb{R}^3$ with positive mesh widths. For a locally integrable vector field, define the cell-observation map with sequence codomain

$$A_h : L_{\text{loc}}^1(\mathbb{R}^3; \mathbb{R}^3) \longrightarrow (\mathbb{R}^3)^{\mathcal{T}_h}, \quad (A_h z)_C = \frac{1}{|C|} \int_C z(x) \, dx.$$

Only coordinatewise equality in this codomain is needed. If a sequence norm is desired for $z \in L^2$, the natural volume-weighted norm obeys $\sum_{C \in \mathcal{T}_h} |C| |(A_h z)_C|^2 \leq \|z\|_2^2$ by Cauchy–Schwarz in each cell and summation. Each grid has infinitely many cells. The next result concerns a finite number of such grids and equality on every cell of each grid, at every time before blowup.

Theorem 4.7 (Identical whole-space cell observations). *Under the regular-reference hypotheses of Theorem 4.2, fix a finite family of complete uniform Cartesian grids. The inserted solutions may be chosen so that, for every grid in that family,*

$$A_h u_\varepsilon(t) = A_h v(t), \quad A_h g_\varepsilon(t) = A_h g(t) \quad (0 \leq t < T),$$

while $T_{\max, \mathbb{R}}^\nu(a, g_\varepsilon) = T$ and the energy and force convergences in Proposition 4.6 hold. All differences are supported inside a ball contained in one cell of every grid, except for an optional spatially constant pressure gauge.

Proof. The union of faces in one grid is a closed, locally finite union of planes. For the fixed finite family their union is closed and has measure zero. Choose a point outside that union and a ball with closure inside one cell of each grid. Apply Theorem 4.2 inside that ball, using the compact pressure representative. Put $\delta u = u_\varepsilon - v$, $\delta p = p_\varepsilon - \pi = P_\varepsilon$, and $\delta g = g_\varepsilon - g$.

Fix one containing cell C . For every $t < T$, δu is smooth, divergence free and compactly supported in its interior. For component j ,

$$\delta u_j = \nabla \cdot (x_j \delta u), \quad \int_C \delta u_j \, dx = \int_{\partial C} x_j \delta u \cdot n \, dS = 0.$$

Subtract the two momentum equations and integrate to obtain

$$\int_C \delta g \, dx = \frac{d}{dt} \int_C \delta u \, dx + \int_{\partial C} (u_\varepsilon \otimes u_\varepsilon - v \otimes v + \delta p I - \nu \nabla \delta u) n \, dS. \quad (53)$$

The time derivative is zero by the preceding identity. All differences vanish in a neighborhood of the cell boundary, so the surface term is zero. A spatially constant pressure gauge adds a multiple of $\int_{\partial C} n \, dS = 0$. The averages therefore agree in C ; in every other cell they agree by support. The same construction fits the containing cell of each grid, and the convergence estimates are unaffected by this fixed upper bound on ε . $\square$

A deterministic procedure using only the initial velocity cell averages and the force cell averages for $0 \leq t < T$ receives identical data in the two cases. Other observations, such as point values or pressure, may distinguish them. The perturbation depends on the fixed grid family and develops larger amplitudes and derivatives as its support shrinks. The result therefore concerns the information in the prescribed averages, rather than convergence under refinement for one fixed smooth problem.

5 Conclusion

Compact support allows the singular solution to be placed inside a small region, and parabolic scaling makes its force small below the critical orders. A cutoff of a local vector potential removes the reference velocity in that region while preserving incompressibility. The resulting exact construction proves density for every fixed admissible smooth initial velocity in the force topologies specified above.

For zero initial velocity, the converse follows from regularity estimates. On both $\mathbb{T}^3$ and $\mathbb{R}^3$, sufficiently small $L_t^1 H_x^{1/2}$ force gives global regularity. On $\mathbb{R}^3$, sufficiently small $L_t^2 H_x^{-1/2}$ force gives regularity on each prescribed finite interval; the permitted radius depends on that interval. These estimates give the sharp thresholds in Theorems 3.1 and 4.1. The whole-space proof controls low frequencies separately, both when scaling negative Sobolev norms and when obtaining the H^2 integrability needed for continuation.

The approximation preserves more than the initial velocity. Near every regular reference, it preserves the earlier history and converges strongly in energy and dissipation. On $\mathbb{R}^3$ it also preserves all velocity and force cell averages for every $0 \leq t < T$ on any fixed finite family of Cartesian grids, when the insertion is placed inside a common cell. The periodic and bounded-domain constructions give prescribed finite collections of singular regions. The whole-space compact solution also admits the infinite-dimensional affine family of velocities in Proposition 3.15, with corresponding changes in the force.

The quantifiers and topologies determine the scope of these conclusions. The force may depend on the initial velocity; no classification of singular initial velocities for a single prescribed force follows. Exact blowup at T is obtained near references regular through T , whereas the general density statement concerns breakdown by T . Small integrated force norms coexist with increasing force amplitudes and derivative norms. Thus density alone gives neither a probability of blowup nor stability under arbitrary perturbations, and it does not imply density within a fixed spectral truncation or a prescribed finite-dimensional force family.

Finally, the energy estimates concern the approach to the singular time. They do not select a continuation afterward or assert nonexistence or nonuniqueness of Leray–Hopf continuations. The grid statement concerns specified observations; the construction is not a computational fluid dynamics experiment.

A Classical local theory and continuation

We derive Proposition 2.1 from the classical forced local theory in Tao [11, Theorems 5.1 and 5.4]. The reductions below specify the viscosity, spatial mean, pressure, and continuation conventions needed here. These results are independent of the compact blowup input.

Lemma A.1 (Sobolev multiplication and embeddings). *For each integer $m \geq 2$ and either domain,*

$$\|vw\|_{H^m} \leq C_m(\|v\|_{H^2} \|w\|_{H^m} + \|w\|_{H^2} \|v\|_{H^m}), \quad \|v\|_\infty \leq C \|v\|_{H^2}. \quad (54)$$

In particular, for integers $k \geq 3$,

$$\|vw\|_{H^k} \leq C_k \|v\|_{H^k} \|w\|_{H^k}, \quad (55)$$

$$\|u \otimes u\|_{H^k} \leq C_k \|u\|_{H^2} \|u\|_{H^k}. \quad (56)$$

Also $H^1 \hookrightarrow L^6$. The homogeneous estimates used in the critical energy arguments are

$$\|v\|_3 \leq C \|v\|_{\dot{H}^{1/2}}, \quad \|\nabla v\|_3 + \|\Lambda v\|_3 \leq C \|v\|_{\dot{H}^{3/2}}, \quad (57)$$

with mean-zero fields on the torus.

Proof. These are standard Sobolev estimates; for the Euclidean product bound see Tao [10, Appendix A, Lemma A.8], together with the H^2 embedding into L^∞ . We recall a short Fourier argument that also covers the torus. The elementary weight inequality $\langle \xi \rangle^m \leq C_m(\langle \eta \rangle^m + \langle \xi - \eta \rangle^m)$ and the Fourier convolution formula give

$$\|vw\|_{H^m} \leq C_m(\|\widehat{v}\|_1 \|w\|_{H^m} + \|\widehat{w}\|_1 \|v\|_{H^m}).$$

Here the Fourier norm is L^1 in the whole space and ℓ^1 on the torus. The convolution bound follows by the triangle inequality for a sum or integral of translates in L^2 . Cauchy–Schwarz gives $\|\widehat{v}\|_1 \leq C \|v\|_{H^2}$, since $\int_{\mathbb{R}^3} \langle \xi \rangle^{-4} d\xi < \infty$ and $\sum_{k \in \mathbb{Z}^3} \langle 2\pi k \rangle^{-4} < \infty$. The latter follows by grouping lattice points in dyadic shells; the shell of radius 2^j contributes at most $C2^{-j}$. This proves the product bounds, first for Schwartz functions or Fourier polynomials and then by approximation. Absolute Fourier convergence also proves the L^∞ estimate. The argument applies componentwise to vectors and tensors, and factorization gives the corresponding product difference bound.

Lemma B.1 proves (57) on both domains. Its order-one case gives $H^1 \hookrightarrow L^6$ on $\mathbb{R}^3$. On the torus apply it to $v - \widehat{v}(0)$ and use $|\widehat{v}(0)| \leq \|v\|_2$. $\square$

Proof of Proposition 2.1. Application of the classical local theory. Tao [11, Theorem 5.1(ii)–(iv)] gives local existence, uniqueness, and smoothness for the forced periodic equation; [11, Theorem 5.4(ii)–(iv)] gives the whole-space counterpart. On each finite interval our force is bounded into H^1 and integrable into every H^m , so it satisfies the source hypotheses. The higher-order bounds in part (ii) hold on the same local interval for every order. Moreover, the proof of Theorem 5.4(iv) explicitly allows $a \in H^k$ and $f \in C_t^j H_x^k$ for all j, k , rather than requiring Schwartz data. This gives the common-interval smoothness used here. More explicitly, the bounds in every spatial Sobolev order and the projected equation first give $u \in W_t^{1,\infty} H_x^k$ for each k , hence continuity into every H^k . The equation then has a continuous right-hand side in every H^k ; repeated time differentiation gives $C_t^j H_x^k$ regularity for all j, k , including one-sided derivatives at the initial time. We apply the projected formulation with force $\mathbb{P}f$; the pressure formulas in Section 2.3 restore the gradient part of f .

The cited theory uses viscosity one. For general $\nu > 0$, set

$$\tau = \nu t, \quad \widetilde{u}(\tau, x) = \nu^{-1} u(\tau/\nu, x), \quad \widetilde{p}(\tau, x) = \nu^{-2} p(\tau/\nu, x), \quad \widetilde{f}(\tau, x) = \nu^{-2} f(\tau/\nu, x).$$

These fields satisfy the equation with viscosity one and preserve the stated smoothness and Sobolev classes. Undoing this change restores ν .

The periodic theorem is stated for mean-zero data. To remove this restriction, define the known functions

$$m(t) = \int_{\mathbb{T}^3} a \, dx + \int_0^t \int_{\mathbb{T}^3} f(r, x) \, dx \, dr, \quad X(t) = \int_0^t m(r) \, dr.$$

The change of variables

$$v(t, x) = u(t, x + X(t)) - m(t), \quad h(t, x) = f(t, x + X(t)) - m'(t)$$

reduces the equation to mean-zero initial velocity and force, with pressure $p(t, x + X(t))$. Indeed, the extra transport from $X' = m$ cancels the term subtracted from the advecting velocity. Spatial translations preserve every Sobolev norm. Solving the reduced problem and reversing the transformation therefore gives the periodic solution with the required pressure normalization; compare [11, Section 3, equations (36)–(38)]. No mean reduction is needed on $\mathbb{R}^3$.

In either domain, the resulting velocity satisfies the mild equation

$$\begin{aligned} u(t) &= e^{\nu t \Delta} a - \int_0^t e^{\nu(t-r)\Delta} \mathbb{P} \nabla \cdot (u \otimes u)(r) \, dr \\ &\quad + \int_0^t e^{\nu(t-r)\Delta} \mathbb{P} f(r) \, dr. \end{aligned} \tag{58}$$

Conversely, smooth solutions in our classes satisfy this equation, using the prescribed pressure gradient and the heat semigroup. For completeness, if $z = u_1 - u_2$ for two such solutions with the same data,

$$\frac{1}{2}(\|z\|_2^2)' + \nu \|\nabla z\|_2^2 \leq \|\nabla u_2\|_\infty \|z\|_2^2.$$

The coefficient is bounded on compact common intervals, since $u_2 \in C_t H^3$. Grönwall gives uniqueness. Patching these local solutions defines the maximal lifespan, as in [11, Corollary 5.2] and the corresponding whole-space argument.

The stated continuation criterion. We retain the short energy argument that yields precisely (14). For every integer $m \geq 3$, pairing the equation with u in H^m , integrating by parts, and using (54) gives

$$\frac{1}{2} \frac{d}{dt} \|u\|_{H^m}^2 + \nu \|\nabla u\|_{H^m}^2 \leq C_m \|u\|_{H^2} \|u\|_{H^m} \|\nabla u\|_{H^m} + \|f\|_{H^m} \|u\|_{H^m}. \tag{59}$$

The pressure term vanishes by solenoidality. These identities are justified by Fourier approximation on compact intervals of smooth existence. Young's inequality and division by the regularized norm $(\|u\|_{H^m}^2 + \zeta^2)^{1/2}$, followed by $\zeta \downarrow 0$, imply

$$(\|u\|_{H^m})' \leq C_{m,\nu} \|u\|_{H^2}^2 \|u\|_{H^m} + \|f\|_{H^m}. \tag{60}$$

If (14) holds, Grönwall bounds every H^m norm uniformly up to S . The H^1 local existence bounds of the cited Theorems 5.1(ii) and 5.4(ii) then give a common positive existence duration when restarting at $t_0 \uparrow S$: the initial H^1 norms stay bounded, and f is bounded into H^1 on $[0, S + 1]$. Their higher-order regularity statements give smooth solutions on these same intervals. One interval extends beyond S , and uniqueness identifies it with the original solution on the overlap. The periodic mean stays bounded by its displayed formula, so the same restart argument applies there. This proves the asserted continuation without requiring zero mean or a whole-space Poincaré inequality. $\square$

B Classical Sobolev embeddings and their realizations

The estimates below are consequences of classical Sobolev embedding. For the Euclidean homogeneous inequality we use Tao [10, Appendix A, equation (A.11)]; for compact manifolds we use Taylor [14, Chapter 13, Proposition 6.4 and the preceding discussion]. We retain the realization and normalization details needed for our applications. Throughout, $\Lambda = (-\Delta)^{1/2}$ has symbol $|\xi|$ on $\mathbb{R}^3$ and $2\pi|k|$ on the unit torus.

Lemma B.1 (Critical embeddings on the two domains). *For $a \in \{1/2, 1\}$ and $p_a = 6/(3 - 2a)$, there are finite constants, depending only on a and on the fixed unit torus when relevant, such that*

$$\|v\|_{L^{p_a}(\mathbb{R}^3)} \leq C_a \|\Lambda^a v\|_{L^2(\mathbb{R}^3)}, \quad \|v\|_{L^{p_a}(\mathbb{T}^3)} \leq C'_a \|\Lambda^a v\|_{L^2(\mathbb{T}^3)}. \quad (61)$$

The first inequality holds initially for Schwartz functions and extends to the homogeneous completion specified below. The second holds for mean-zero periodic distributions with finite displayed right-hand side; the mean-zero condition is essential. Componentwise application gives the same statements for vector and tensor fields, with constants allowed to depend on their fixed number of components.

Consequently, on either domain and whenever the indicated homogeneous norms are finite (with mean-zero representatives on the torus),

$$\begin{aligned} \|v\|_3 &\leq C \|v\|_{\dot{H}^{1/2}}, \\ \|\nabla v\|_3 + \|\Lambda v\|_3 &\leq C \|v\|_{\dot{H}^{3/2}}, \\ \|\nabla v\|_6 &\leq C \|\Delta v\|_2. \end{aligned} \quad (62)$$

Here the derivative statements on $\mathbb{R}^3$ refer to distributions whose derivatives have the homogeneous realizations described in the proof; in particular they apply to all the smooth H^∞ fields used in the article.

Proof. Euclidean embedding and the homogeneous realization. The classical homogeneous Sobolev inequality [10, Appendix A, equation (A.11)], with dimension three, base exponent two, and $0 < a < 3/2$, gives

$$\|v\|_{L^{p_a}(\mathbb{R}^3)} \leq C_a \|\Lambda^a v\|_2, \quad \frac{1}{p_a} = \frac{1}{2} - \frac{a}{3}.$$

Tao's Fourier convention includes the compensating Plancherel factor specified in Section 1, so this is exactly the norm used here. Initially take $\widehat{v} \in C_c^\infty(\mathbb{R}^3 \setminus \{0\})$ and complete this class in $\|\Lambda^a v\|_2$.

We specify its distributional realization to avoid any ambiguity by polynomials. If $G \in L^2$ is the limit of $|\xi|^a \widehat{v}_n$, set $\widehat{v} = |\xi|^{-a} G$. For every Schwartz test function ϕ ,

$$\int_{\mathbb{R}^3} |\xi|^{-a} |G(\xi) \widehat{\phi}(\xi)| \, d\xi \leq \|G\|_2 \left(\int_{\mathbb{R}^3} |\xi|^{-2a} |\widehat{\phi}(\xi)|^2 \, d\xi \right)^{1/2} < \infty.$$

The last integral is finite at zero because $2a < 3$, and at infinity by rapid decay. This estimate also proves distributional convergence. The embedding makes v_n Cauchy in L^{p_a} , whose limit represents the same distribution by Hölder's inequality. Every $G \in L^2$ is approximable by smooth functions supported in compact annuli, by truncation and mollification. Thus this construction gives a unique L^{p_a} representative for every element of the completion. It includes all Schwartz functions and all H^∞ fields with finite displayed homogeneous norm. Taking $a = 1/2$ and $a = 1$ gives $p_a = 3$ and $p_a = 6$, respectively.

The unit torus. The compact-manifold Sobolev embedding in [14, Chapter 13, Proposition 6.4 and the preceding discussion] yields

$$\|v\|_{L^{p_a}(\mathbb{T}^3)} \leq C_a \|v\|_{H^a(\mathbb{T}^3)} \quad (a = 1/2, 1).$$

Taylor's angular torus coordinates are converted to the unit torus as explained in Section 1; the resulting inhomogeneous norms are equivalent. For mean-zero fields the spectral gap gives

$$(1 + 4\pi^2 |k|^2)^a \leq (1 + (2\pi)^{-2})^a (2\pi |k|)^{2a}, \quad k \neq 0,$$

and hence $\|v\|_{H^a(\mathbb{T}^3)} \leq C'_a \|v\|_{\dot{H}^a(\mathbb{T}^3)}$. This proves the periodic inequality. Fourier truncations converge in $\dot{H}^a$ for every mean-zero periodic distribution with finite norm, and therefore also in L^{p_a} ; their distributional limits identify the representatives.

Derivative estimates. Apply the $a = 1/2$ inequality to each $\partial_j v$ and to Λv , and the $a = 1$ inequality to each $\partial_j v$. Plancherel and $\sum_j |\xi_j|^2 = |\xi|^2$, or its Fourier-series counterpart, give (62). The Euclidean statements use the preceding realizations of the derivatives; no embedding of $\dot{H}^{3/2}$ into L^∞ is asserted. Componentwise application covers vectors and tensors. In particular, for smooth fields,

$$| \langle (v \cdot \nabla) v, \Lambda v \rangle | \leq \|v\|_3 \|\nabla v\|_3 \|\Lambda v\|_3 \leq C \|v\|_{\dot{H}^{1/2}} \|v\|_{\dot{H}^{3/2}}^2.$$

All constants depend only on the fixed exponents, domain, and norm conventions, not on the field or its frequency support. $\square$

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